DIFFERENTIAL GEOMETRY AND LIE GROUPS

Jishnu Biswas, notes by Ramdas Singh

Fifth Semester

Contents

1 PRELIMINARIES AND MANIFOLDS 1
- 1.1 Calculus on $\mathbb{R}^n$ . . . 1
- 1.2 Multilinear Algebra . . . 3
  - 1.2.1 The Tensor and Wedge Products . . . 4
- 1.3 Building Blocks for Manifolds . . . 6
  - 1.3.1 Exterior Differentiation . . . 8
- 1.4 Manifolds . . . 9
  - 1.4.1 Smooth Functions and Smooth Maps . . . 11
  - 1.4.2 Quotient Manifolds . . . 13
- 1.5 Differential Structure . . . 15
- 1.6 Regular Maps and Regular Submanifolds . . . 17

2 TANGENT SPACES, BUNDLES, AND VECTORS 21
- 2.1 Manifold Structure . . . 21
  - 2.1.1 Sections and Frames . . . 23
- 2.2 Bump Functions and Partitions of Unity . . . 24
- 2.3 Integral Curves . . . 26
- 2.4 Lie Algebras and Lie Groups . . . 28
  - 2.4.1 Matrix Lie Groups and Lie Algebras . . . 29

3 FORMS, ORIENTATION, AND INTEGRATION 35
- 3.1 Differential Forms . . . 35
  - 3.1.1 k-forms . . . 37
  - 3.1.2 Exterior Differentiation . . . 39
- 3.2 Orientable Manifolds . . . 40
  - 3.2.1 Manifolds with Boundary . . . 42

Index 45

Chapter 1

PRELIMINARIES AND MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds*. Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

1.1 Calculus on $\mathbb{R}^n$

A point p or element in $\mathbb{R}^n$ is associated with an n-tuple of coordinates $x^1, x^2, \ldots, x^n$. Let U be an open neighbourhood of p in $\mathbb{R}^n$. For a non-negative integer k, we say that a function $f : U \to \mathbb{R}$ is C^k*-continuous* at $p \in U$ if all partial derivatives of f of order $\leq k$ exist and are continuous at p. f is said to be C^∞ or *smooth* if f is C^k for all $k \geq 0$. f is C^k (or C^∞)on U if it is C^k (or C^∞) at every point in U.

If f happens to be a vector-valued function, i.e. $f : U \to \mathbb{R}^m$, we say $f = (f^1, f^2, \ldots, f^m)$ is C^k (or C^∞) at p if each component function $f^i : U \to \mathbb{R}$ is C^k (or C^∞) at p. f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U.

Definition 1.1. Let R be a commutative ring with unity. We term A to be an R*-algebra* if A is a commutative ring with unity and an R-module, and for $r \in R$ and $a, b \in A$, we have $r(ab) = (ra)b = a(rb)$.

In this course, we will be primarily be concerned with the case when R is a field, and A is an R-vector space.

For a subset $U \subseteq \mathbb{R}^n$, we denote $C^k(U)$ (or $C^\infty(U)$) as the set of all C^k (or C^∞) functions from U to $\mathbb{R}$. Note that this is a commutative ring with unity. It is also a vector space over $\mathbb{R}$. Together, $C^k(U)$ (or $C^\infty(U)$) is a (commutative) algebra over $\mathbb{R}$. Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \cdots C^k(U) \supseteq C^{k+1}(U) \supseteq \cdots \tag{1.1}$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \tag{1.2}$$

$f : U \to \mathbb{R}$ is termed *real analytic* at $p \in U$ if there exists a neighbourhood $V \subseteq U$ of p such that

$$f(x) = f(p) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \cdots + \frac{1}{k!} \sum_{1 \leq i_1, \ldots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \cdots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \cdots (x^{i_k} - p^{i_k}) + \cdots \tag{1.3}$$

for all $x \in V$. f is said to be real analytic on U if it is real analytic at every point in U. We denote the set of all real analytic functions on U as $C^\omega(U)$. Note that $C^\omega(U) \subseteq C^\infty(U)$.

Theorem 1.2 (*Taylor's theorem*)**.** *Let $p \in U \subseteq \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be a C^∞ function. Further assume that for all $x \in U$, the line segment from p to x is contained in U (U in this case is said to be* star-shaped *with respect to p). Then there exist $g_1, \ldots, g_n \in C^\infty(U)$ such that, for all $x \in U$,*

$$f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i), \tag{1.4}$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \leq i \leq n. \tag{1.5}$$

Proof. Let $x \in U$. Then the line segment $\{p + t(x-p) \mid t \in [0,1]\}$ is contained in U. Composing f with the map $t \mapsto p + t(x-p)$, we obtain

$$\frac{df}{dt}(p + t(x-p)) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p + t(x-p))(x^i - p^i). \tag{1.6}$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)\int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)g_i(x). \tag{1.7}$$

Clearly, because f is C^∞, each g_i is also C^∞. Further, we have $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$. ∎

Let $p = (p^1, \ldots, p^n), v \in \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be C^∞ at p. The *directional derivative* of$f$ at p in the direction of v is defined as

$$\lim_{t\to 0} \frac{f(p+tv) - f(p)}{t} = \frac{d}{dt} f(p+tv)\Big|_{t=0} = \sum_{i=1}^{n} \frac{df}{dx^i}(p)v^i. \tag{1.8}$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\Big|_p \tag{1.9}$$

Now consider all pairs of the form (f, U) where $p \in U \subseteq \mathbb{R}^n$ is an open neighbourhood of p and $f : U \to \mathbb{R}$ is C^∞. Define an equivalence relation $\sim$ on this set by declaring $(f, U) \sim (g, V)$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) is denoted by $[(f, U)]$, and the set of all such equivalence classes is denoted by $C_p^\infty(\mathbb{R}^n)$. Such an equivalence class $[(f, U)]$ is called the *germ* of f at p. We can give $C_p^\infty(\mathbb{R}^n)$ the structure of a commutative ring with unity by defining

$$[(f,U)] + [(g,V)] = [(f+g, U\cap V)], \quad [(f,U)]\cdot[(g,V)] = [(fg, U\cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \tag{1.10}$$

It can also be given the structure of a vector space over $\mathbb{R}$ by defining

$$\lambda[(f,U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \tag{1.11}$$

Thus, $C_p^\infty(\mathbb{R}^n)$ is an $\mathbb{R}$-algebra. Now, given a germ $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$ and a vector $v \in \mathbb{R}^n$, we define

$$D_v\alpha = \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}\Big|_p = v \cdot \nabla f(p). \tag{1.12}$$

Note that the choice of f is immaterial, since if $[(f, U)] = [(g, V)]$, then there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$, and hence $\nabla f(p) = \nabla g(p)$. One can infer that $D_{\lambda v}(f) = \lambda D_v(f) =$

$D_v(\lambda f)$ and $D_v(f+g) = D_v(f) + D_v(g)$ for all $\lambda \in \mathbb{R}$ and $f, g \in C_p^\infty(\mathbb{R}^n)$. Thus, D_v is an $\mathbb{R}$-linear map from $C_p^\infty(\mathbb{R}^n)$ to $\mathbb{R}$ (hereforth, we shall use f to also denote the germ $[(f,U)]$ of f at p). Also,

$$D_v(fg) = \sum_{i=1}^n v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left(f(p)\frac{\partial g}{\partial x^i}(p) + g(p)\frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \tag{1.13}$$

Such a property makes D_v what we call a derivation.

Definition 1.3. Any $\mathbb{R}$-linear map $T : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}$ satisfying the Leibniz rule $T(fg) = f(p)T(g) + g(p)T(f)$ for all $f, g \in C_p^\infty(\mathbb{R}^n)$ is called a *derivation* at p.

One may show that if T is a derivation at p, then $T(1) = 0$.

The set of all derivations at p is denoted by $\mathfrak{D}_p(\mathbb{R}^n)$. It is easy to see that $\mathfrak{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. Now consider the map

$$\theta : \mathbb{R}^n \to \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \tag{1.14}$$

This θ is an $\mathbb{R}$-linear map (verify). We claim that θ is an isomorphism of real vector spaces. To this end, let $v \in \mathbb{R}^n$ such that $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$. Then $D_v f = 0$ for all $f \in C_p^\infty(\mathbb{R}^n)$. In particular, for the coordinate functions x^i, we have $D_v x^i = v_i = 0$ for all $1 \le i \le n$. Thus, $v = 0$, and hence θ is injective. To show surjectivity, we appeal to Taylor's theorem; let $D \in \mathfrak{D}_p(\mathbb{R}^n)$. We claim that $D = D_v$ where $v = (D(x^1), \dots, D(x^n))$. Indeed, for any $f \in C_p^\infty(\mathbb{R}^n)$, we have

$$\begin{aligned} D(f) &= D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i) g_i(x) \right) = 0 + \sum_{i=1}^n \left((x^i - p^i)(p) Dg_i(x) + D(x^i - p^i) g_i(p) \right) \\ &= \sum_{i=1}^n (Dx^i) g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p) Dx^i = D_v(f). \end{aligned} \tag{1.15}$$

Hence, θ is surjective, and we have the desired isomorphism $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$.

Note that a germ $f \in C_p^\infty(\mathbb{R}^n)$ has a valid value at p, but no other point. Hence, we have the evaluation map

$$\mathrm{ev}_p : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}, \quad [(f,U)] \mapsto f(p). \tag{1.16}$$

The kernel here is $\{[(f,U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$, which is a ideal in $C_p^\infty(\mathbb{R}^n)$. For any other element $[(f,U)] \in C_p^\infty(\mathbb{R}^n)$ with $f(p) \neq 0$, a valid inverse is given by $[(f^{-1}, U \setminus \{f = 0\})]$. This makes $C_p^\infty(\mathbb{R}^n)$ a local ring.

1.2 Multilinear Algebra

July 22nd.

Hereforth, V shall denote a finite-dimensional real vector space, and V^V shall denote its *dual space* $\mathrm{Hom}_{\mathbb{R}}(V, \mathbb{R})$. If $e_1, \dots, e_n$ are a basis for V, then the *dual basis* of V^V is denoted by $\alpha^1, \dots, \alpha^n \in V^V$, and is defined by $\alpha^i(e_j) = \delta_{i,j}$ for all $1 \le i, j \le n$. One may verify that the dual basis $\{\alpha^1, \dots, \alpha^n\}$, induced by the basis $\{e_1, \dots, e_n\}$, is indeed a basis for V^V. S_k shall denote the permutation group on k symbols, satisfying $|S_k| = k!$. For any $\sigma \in S_k$, the sign of σ is denoted by $\mathrm{sgn}(\sigma)$, and is defined as $+1$ if σ is an even permutation, and -1 if σ is an odd permutation.

With notation out of the way, let $k \ge 1$ be an integer. We shall consider the k-tuple $V^k = V \times \cdots \times V$ (k times).

Definition 1.4. A mapping $f : V^k \to \mathbb{R}$ is said to be *k-linear* if f is linear in each of its k arguments.

A special case is when $k = 2$, in which case f is called a *bilinear form*. Such an $f : V \times V \to \mathbb{R}$ satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v,u) + \mu f(w,u), \quad f(u, \lambda v + \mu w) = \lambda f(u,v) + \mu f(u,w). \tag{1.17}$$

A common example of a bilinear form is any valid inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ on V. Another example is the determinant $\det : \mathbb{R}^n \to \mathbb{R}$, where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \to \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \tag{1.18}$$

It is easy to see that $L_k(V)$ is an $\mathbb{R}$-vector space.

Definition 1.5. A k-linear map $f : V^k \to \mathbb{R}$ is called a *symmetric k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.19}$$

That is, f is invariant under any permutation of its arguments. Moreover, f is called an *alternating k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \ldots, v_k) = \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.20}$$

One can see that the standard inner product on $\mathbb{R}^n$ is a symmetric bilinear form, and the determinant is an alternating n-linear form. Related to these definitions, we can immediately define the following subspaces of $L_k(V)$:

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \tag{1.21}$$

Currently, our main interest of study is $A_k(V)$, the space of alternating k-linear forms on V. Elements of $A_k(V)$ are termed *alternating k-tensors*. Similarly, those of $S_k(V)$ and $L_k(V)$ are called *symmetric k-tensors* and *k-tensors*, respectively. For $k = 0$, we define $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$. For $k = 1$, we have $L_1(V) = S_1(V) = A_1(V) = V^V$. We can define the (left) action of S_k on $L_k(V)$ by

$$(\sigma \cdot f)(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.22}$$

Thus, f is symmetric if $\sigma \cdot f = f$ for all $\sigma \in S_k$, and f is alternating if $\sigma \cdot f = \operatorname{sgn}(\sigma) f$ for all $\sigma \in S_k$.

Given an $f \in L_k(V)$, we can define the *symmetrization* $Sf \in S_k(V)$, of f, as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \tag{1.23}$$

One may, indeed, check that Sf is symmetric. Similarly, we can define the *alternation* $Af \in A_k(V)$, of f, as

$$Af = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) \sigma \cdot f. \tag{1.24}$$

1.2.1 The Tensor and Wedge Products

July 24th.

For any $f \in L_k(V)$ and $g \in L_l(V)$, we can define the *tensor product* $f \otimes g \in L_{k+l}(V)$ as

$$(f \otimes g)(v_1, \ldots, v_{k+l}) = f(v_1, \ldots, v_k) g(v_{k+1}, \ldots, v_{k+l}). \tag{1.25}$$

Verify that $f \otimes g$ is indeed $(k+l)$-linear. We are interested in the tensor product of alternating tensors, so let $f \in A_k(V)$ and $g \in A_l(V)$. Then, we can define the *wedge product* $f \wedge g \in A_{k+l}(V)$ as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot (f \otimes g). \tag{1.26}$$

The associativity of the tensor product holds easily: for $f \in L_k(V)$, $g \in L_l(V)$, and $h \in L_m(V)$,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \tag{1.27}$$

Proposition 1.6. *For $f \in A_k(V)$ and $g \in A_l(V)$, we have*

$$f \wedge g = (-1)^{kl} g \wedge f. \tag{1.28}$$

Proof. We shall show that $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$. To this end, fix the permutation $\tau \in S_{k+l}$ defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \tag{1.29}$$

We have

$$\begin{aligned} A(f \otimes g)(v_1, \ldots, v_{k+l}) &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+l)}) \\ &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma\tau(l+1)}, \ldots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \ldots, v_{\sigma\tau(l)}) \\ &= \operatorname{sgn}(\tau) \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)}, \ldots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \ldots, v_{\sigma\tau(l)}) \\ &= \operatorname{sgn}(\tau) A(g \otimes f)(v_1, \ldots, v_{k+l}). \end{aligned} \tag{1.30}$$

With $\operatorname{sgn}(\tau) = (-1)^{kl}$, the proof is complete. ∎

Corollary 1.7. *For $f \in A_k(V)$, we have $f \wedge f = 0$ if k is odd.*

Proof. Easy since $f \wedge f = (-1)^{k^2} f \wedge f = -f \wedge f$ if k is odd, so $f \wedge f = 0$. ∎

Lemma 1.8. *For $f \in L_k(V)$ and $g \in L_l(V)$, we have*

$$A(Af \otimes g) = k! A(f \otimes g), \quad A(f \otimes Ag) = l! A(f \otimes g). \tag{1.31}$$

Proof. We shall prove the first equality; the second follows similarly. We have

$$\begin{aligned} A(Af \otimes g) &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \operatorname{sgn}(\tau)(\tau \cdot f) \otimes g \right) = \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \operatorname{sgn}(\tau) \tau \cdot (f \otimes g) \right) \\ &= \sum_{\tau \in S_k} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma\tau) \sigma\tau \cdot (f \otimes g) = k! \sum_{\alpha \in S_{k+l}} \operatorname{sgn}(\alpha) \alpha \cdot (f \otimes g) = k! A(f \otimes g). \end{aligned} \tag{1.32}$$

∎

Proposition 1.9. *The wedge product is associative, i.e. for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$, we have*

$$(f \wedge g) \wedge h = f \wedge (g \wedge h). \tag{1.33}$$

Proof. We have

$$\begin{aligned} (f \wedge g) \wedge h &= \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)!m!k!l!} A(A(f \otimes g) \otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!} A((f \otimes g) \otimes h) \\ &= \frac{1}{k!l!m!} A(f \otimes g \otimes h) = f \wedge (g \wedge h). \end{aligned} \tag{1.34}$$

∎

Corollary 1.10. *$f \wedge g \wedge h = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$ for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$. Inductively,*

$$f_1 \wedge \cdots \wedge f_k = \frac{1}{k_1! \cdots k_k!} A(f_1 \otimes \cdots \otimes f_k), \tag{1.35}$$

where $f_i \in A_{k_i}(V)$ for $1 \leq i \leq k$.

Proposition 1.11. *Let $\alpha^1, \ldots, \alpha^n \in V^V = A_1(V) = L_1(V)$, and $v_1, \ldots, v_k \in V$. Then*

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}. \tag{1.36}$$

Proof. We have

$$\begin{aligned}(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) &= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)(\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\alpha^1(v_{\sigma(1)})\alpha^2(v\sigma(2)) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}. \end{aligned} \tag{1.37}$$

■

July 27th.

Let V be an $\mathbb{R}$-vector space with basis $e_1, \ldots, e_n$, and V^V be its dual space with dual basis $\alpha^1, \ldots, \alpha^n$. Moreover, let I be a multi-index of length k, i.e. $I = (i_1, \ldots, i_k)$ with $1 \leq i_1, \ldots, i_k \leq n$. We define $e_I := (e_{i_1}, \ldots, e_{i_k}) \in V^k$ and $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$. Then, a k-linear form $f \in L_k(V)$ is completely determined by its values on the e_I, and an alternating k-linear form $f \in A_k(V)$ is completely determined by its values on the e_I with $i_1 < i_2 < \cdots < i_k$.

To this end, we first claim that $\alpha^I(e_J) = \delta^I_J$ for all multi-indices I, J of length k. Indeed, if $J = (j_1, \ldots, j_k)$ with $j_1 < j_2 < \cdots < j_k$, then

$$\alpha^I(e_J) = \det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k}; \tag{1.38}$$

. this is 1 for $I = J$. Suppose $I \neq J$. Let $1 \leq l \leq k$ be the first index where $i_l \neq j_l$. Then the lth row of $\alpha^{i_m}(e_{j_n})_{1 \leq m,n \leq k}$ is an all zeroes row, and hence $\det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k} = 0$. Thus, $\alpha^I(e_J) = \delta^I_J$. We now claim that α^I, for strictly increasing multi-indices I of length k, form a basis for $A_k(V)$. For linear independence, we simply have (the sums in this context are over strictly increasing multi-indices I of length k):

$$\sum_I c_I \alpha^I = 0 \implies \sum_I c_I \alpha^I(e_J) = 0 \implies c_J = 0, \quad \forall J. \tag{1.39}$$

To show that they span $A_k(V)$, let $f \in A_k(V)$. Then consider the k-linear form

$$g = \sum_I f(e_I)\alpha^I. \tag{1.40}$$

We show that $f = g$; it is enough to prove that $f(e_J) = g(e_J)$ for all strictly increasing multi-indices J of length k. We have

$$g(e_J) = \sum_I f(e_I)\alpha^I(e_J) = f(e_J)\alpha^J(e_J) + \sum_{I \neq J} f(e_I)\alpha^I(e_J) = f(e_J). \tag{1.41}$$

Corollary 1.12. *$A_k(V) = 0$ if $k > n$. Further, $\dim A_k(V) = \binom{n}{k}$ if $1 \leq k \leq n$.*

1.3 Building Blocks for Manifolds

Recall that we had defined the class of germs $C^\infty_p(\mathbb{R}^n)$ of C^∞ functions at a point $p \in \mathbb{R}^n$, which agree on some neighbourhood of p. One can show that $C^\infty_p(\mathbb{R}^n)$ and $C^\infty_p(U)$ are isomorphic as $\mathbb{R}$-algebras, where $U \subseteq \mathbb{R}^n$ is an open neighbourhood of p. Note that we term $T_p(U) = T_p(\mathbb{R}^n) = \mathbb{R}^n$ as the *tangent space* of U (or $\mathbb{R}$) at p.

Definition 1.13. Let $U \subseteq \mathbb{R}^n$ be an open subset. A *vector field* on U is a function $X : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)$ such that $X(p) \in T_p(\mathbb{R}^n)$ for all $p \in U$. In particular,

$$X(p) = \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p, \quad a^i(p) \in \mathbb{R}. \tag{1.42}$$

We can rewrite this as

$$X = \sum_{i=1}^{n} a^i \frac{\partial}{\partial x^i}, \quad a^i : U \to \mathbb{R}. \tag{1.43}$$

X is called a *smooth vector field* if each a^i is a smooth function on U. The set of all smooth vector fields on U is denoted by $\mathfrak{X}(U)$.

This is a real vector space; note that for $X, Y \in \mathfrak{X}(U)$, we have $X + Y \in \mathfrak{X}(U)$, and for $\lambda \in \mathbb{R}$, we have $\lambda X \in \mathfrak{X}(U)$. Further, $\mathfrak{X}(U)$ is a $C^\infty(U)$-module since for $f \in C^\infty(U)$ and $X \in \mathfrak{X}(U)$, we have $fX \in \mathfrak{X}(U)$, where

$$(fX)(p) = f(p)X(p), \quad p \in U. \tag{1.44}$$

Notice that $X \in \mathfrak{X}(U)$ describes a map $C^\infty(U) \to C^\infty(U)$, where $f \mapsto Xf$, being defined as $(Xf)(p) = X_p f$. This is a derivation on $C^\infty(U)$, since for $f, g \in C^\infty(U)$, we have

$$(X(fg))(p) = X_p(fg) = f(p)X_p g + g(p)X_p f = (fXg)(p) + (gXf)(p). \tag{1.45}$$

We refine our definition of a derivation as follows.

Definition 1.14. Let K be a field, and A be a K-algebra. A *derivation* $D : A \to A$ is an $\mathbb{R}$-linear map satisfying the Leibniz rule $D(ab) = aD(b) + bD(a)$ for all $a, b \in A$.

Definition 1.15. A *differential 1-form* on U is a smooth function $\omega : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)^V$ such that $\omega(p) \in T_p(\mathbb{R}^n)^V =: T_p^*(\mathbb{R}^n)$ for all $p \in U$.

For example, for $f \in C^\infty(U)$, we can define the differential 1-form df on U by $df(p) = df_p \in \mathrm{Hom}_{\mathbb{R}}(T_p\mathbb{R}^n, \mathbb{R})$, where $df_p(X) = X_p f$ for all $X \in T_p\mathbb{R}^n$. In particular, dx^i is a differential 1-form on U for all $1 \le i \le n$. In fact, $\{dx^1, \dots, dx^n\}$ is the dual basis of $T_p^*\mathbb{R}^n$ corresponding to the basis $\{\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\}$ of $T_p\mathbb{R}^n$. Thus, a differential 1-form ω on U can be expressed as

$$\omega(p) = \sum_{i=1}^{n} a_i(p) dx^i\Big|_p, \quad a_i : U \to \mathbb{R}. \tag{1.46}$$

$T_p^*(\mathbb{R}^n)$, here, is termed the *cotangent space* of $\mathbb{R}^n$ at p. Thus, our current setup is as follows: $V = T_p\mathbb{R}^n$, and $V^V = T_p^*\mathbb{R}^n$. The basis for $T_p\mathbb{R}^n$ considered is $\{\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\}$, and the corresponding dual basis for $T_p^*\mathbb{R}^n$ is $\{dx^1\Big|_p, \dots, dx^n\Big|_p\}$. And finally, we have

$$A_k(T_p\mathbb{R}^n) = \text{the space of alternating } k\text{-linear forms on } T_p\mathbb{R}^n \tag{1.47}$$

which has the basis $\{dx^{i_1}\Big|_p \wedge \cdots \wedge dx^{i_k}\Big|_p \mid 1 \le i_1 < \cdots < i_k \le n\}$.

Definition 1.16. A *differential k-form* on U is a smooth function $\omega : U \to \amalg_{p\in U} A_k(T_p\mathbb{R}^n)$ such that $\omega(p) \in A_k(T_p\mathbb{R}^n)$ for all $p \in U$.

Any such ω may be written as $\omega = \sum_I a_I dx_I$. It termed a smooth differential k-form if each a_I is a smooth function on U. The set of all smooth differential 1-forms on U is denoted by $\Omega^1(U)$, and the set

of all smooth differential k-forms on U is denoted by $\Omega^k(U)$. Both these sets are $\mathbb{R}$-vector spaces, and $\Omega^k(U)$ is a free $C^\infty(U)$-module of rank $\binom{n}{k}$. The isomorphism may be given as

$$\Omega^1(U) \cong C^\infty(U)^{\oplus\binom{n}{k}}, \quad \omega = \sum_I a_I dx_I \mapsto (a_1, a_2, \ldots, a_{\binom{n}{k}}). \tag{1.48}$$

We can endow $\Omega^k(U)$ with the wedge product $\wedge : \Omega^k(U) \times \Omega^l(U) \to \Omega^{k+l}(U)$, defined as

$$(\omega \wedge \eta)(p) = \omega(p) \wedge \eta(p), \quad p \in U, \tag{1.49}$$

or

$$\omega \wedge \eta = \sum_{I,J} a_I b_J dx_I \wedge dx_J \tag{1.50}$$

where I and J are strictly increasing multi-indices of length k and l, respectively. Note that if $I \cap J \neq \emptyset$, then $dx_I \wedge dx_J = 0$. Thus, the sum may be taken over all strictly increasing multi-indices I and J of length k and l, respectively, such that $I \cap J = \emptyset$. Also,

$$A_*(V) = \bigoplus_{k=0}^{n} A_k(V) \tag{1.51}$$

is defined as the *exterior algebra* of V. The wedge product on here is associative and anti-commutative. We define $\Omega^0(U) = C^\infty(U)$. The map $d : \Omega^k(U) \to \Omega^{k+1}(U)$ is defined as the *exterior derivative*; for $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.52}$$

For a k-form $\omega = \sum_I a_I dx_I \in \Omega^k(U)$, we have

$$d\omega = \sum_I da_I \wedge dx_I = \sum_I \sum_{j=1}^{n} \frac{\partial a_I}{\partial x^j} dx^j \wedge dx_I. \tag{1.53}$$

Example 1.17. Let $U \subseteq \mathbb{R}^3$. Let us call the coordinates $x^1 = x$, $x^2 = y$, and $x^3 = z$. The only non-zero forms are $\Omega^0(U)$, $\Omega^1(U)$, $\Omega^2(U)$, and $\Omega^3(U)$. Note that $\Omega^0(U) = C^\infty(U)$, and there's nothing to say about this. For $\Omega^1(U)$, a form looks like $f\,dx + g\,dy + h\,dz$ for $f, g, h \in C^\infty(U)$. For $\Omega^2(U)$, a form looks like $R\,dx \wedge dy + Q\,dx \wedge dz + P\,dy \wedge dz$ for $f, g, h \in C^\infty(U)$.

July 29th.

The map $\Omega^1(U) \times \mathfrak{X}(U) \to C^\infty(U)$, defined as $(\omega, X) \mapsto \omega(X)$, is $\mathbb{R}$-bilinear. In particular, for $\omega = \sum_{i=1}^n a_i dx^i$ and $X = \sum_{j=1}^n b^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X) = \sum_{i=1}^{n} a_i dx^i \left(\sum_{j=1}^{n} b^j \frac{\partial}{\partial x^j} \right) = \sum_{i=1}^{n} a_i b^i \in C^\infty(U). \tag{1.54}$$

Extending this, we have the map $\Omega^k(U) \times \mathfrak{X}(U)^k \to C^\infty(U)$, defined as $(\omega, X_1, \ldots, X_k) \mapsto \omega(X_1, \ldots, X_k)$, which is $\mathbb{R}$-multilinear. In particular, for $\omega = \sum_I b_I dx^I$ and $X_i = \sum_{i=1}^n a_i^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X_1, \ldots, X_k) = \sum_I b_I a_{i_1}^1 \cdots a_{i_k}^k \in C^\infty(U) \tag{1.55}$$

using the fact that $dx^I(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}) = 1$ if the multi-index matches, and 0 otherwise.

1.3.1 Exterior Differentiation

Here, d_k shall denote the map $d_k : \Omega^k(U) \to \Omega^{k+1}(U)$, defined as the *exterior derivative*. The subscript k is often omitted, and we simply write $d : \Omega^k(U) \to \Omega^{k+1}(U)$. For $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.56}$$

Analogously, for a general k-form $\omega = \sum_I a_I dx^I \in \Omega^k(U)$, we have

$$d : \Omega^k(U) \to \Omega^{k+1}(U), \quad \omega \mapsto d\omega = \sum_I da_I \wedge dx^I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx^I. \tag{1.57}$$

All the d maps are $\mathbb{R}$-linear, but not $C^\infty(U)$-linear. Also, for two forms $\omega \in \Omega^k(U)$ and $\eta \in \Omega^l(U)$, we have

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta. \tag{1.58}$$

This is known as *antiderivation*. Moreover, $d_{k+1} \circ d_k = d^2 = 0$. Both these properties may be verified by the reader. Now suppose we have another map $D : \Omega^k(U) \to \Omega^{k+1}(U)$ which is $\mathbb{R}$-linear, satisfies antiderivation, and $D^2 = 0$, and $Df = df$ for all $f \in C^\infty(U)$. Then $D = d$; the exterior derivative is unique. Thus, a chain of maps is formed:

$$0 \xrightarrow{d} C^\infty(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(U) \xrightarrow{d} 0 \quad \text{for open } U \subseteq \mathbb{R}^n. \tag{1.59}$$

An $\omega \in \Omega^k(U)$ is called an *exact k-form* if $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(U)$. It is called a *closed k-form* if $d\omega = 0$. Note that every exact form is closed, but the converse is not true in general. We will refine the following later: quotienting the space of closed k-forms by the space of exact k-forms gives us the *de Rham cohomology* of U, denoted by $H^k_{\mathrm{dR}}(U)$.

1.4 Manifolds

July 31st.

Definition 1.18. A topological space M is termed a *topological manifold* of dimension n if

1. M is Hausdorff,
2. M is secound countable (that is, has a countable basis), and
3. for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ and a homemorphism $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$, where $\varphi(U)$ is open. This makes a manifold *locally euclidean*.

The *invariance of dimension* property states that if we have two open sets $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ such that U is homeomorphic to V, then $m = n$. Thus, the dimension of a topological manifold is well-defined.

Definition 1.19. A pair (U, φ) is called a *coordinate chart* on M if $U \subseteq M$ is open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.

A trivial example is $\mathbb{R}^n$ with one (coordinate) chart $(\mathbb{R}^n, \mathrm{id}_{\mathbb{R}^n})$. In fact, if $U \subseteq \mathbb{R}^n$ is open, then (U, id_U) is a coordinate chart. Note that subspaces inherit the properties of Hausdorffness and second countability. We state some properties without proof; if X is locally Euclidean but not T_2, then X being a topological manifold implies that the number of connected (or path connected) components is at most countable.

Definition 1.20. Two charts (U, φ) and (V, ψ) on a topological manifold M are said to be C^∞-*compatible* if either

1. $U \cap V = \emptyset$, or
2. $U \cap V \neq \emptyset$, and the transition maps $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ and $\varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$ are C^∞-maps between open subsets of $\mathbb{R}^n$.

Similarly, C^k-compatibility may be defined for $k \geq 0$. A C^ω-manifold is termed to be a real analytic manifold. In case of $k = 2m$, then $\mathbb{R}^{2m}$ is identified with $\mathbb{C}^m$, and the transition maps are required to be holomorphic.

Definition 1.21. M is a C^∞-manifold or a *smooth manifold* if M is a topological manifold and there exists a family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ such that

1. $\{U_\alpha\}_{\alpha\in\Lambda}$ is an open cover of M, and
2. $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are C^∞-compatible for all $\alpha, \beta \in \Lambda$.

Similarly, C^k-manifolds, C^ω-manifolds, and complex manifolds may be defined. The family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ is called an *atlas* of M.

The most basic non-trivial example is perhaps the circle $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$, which is a C^∞-manifold of dimension 1. We can think of $\mathbb{R}^2$ as $\mathbb{C}$. Let $U = S^1 \setminus \{1\} = \{e^{it} \mid 0 < t < 2\pi\}$ and $V = S^1 \setminus \{-1\} = \{e^{it} \mid -\pi < t < \pi\}$. Then (U, φ) and (V, ψ) are charts, where $\varphi : U \to (0, 2\pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$ and $\psi : V \to (-\pi, \pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$. Here, $U \cap V = \{e^{it} \mid 0 < t < \pi\} \cup \{e^{-t} \mid \pi < t < 2\pi, -\pi < t < 0\}$. The transition maps are $\psi \circ \varphi^{-1} : (0, \pi) \to (0, \pi)$ defined as $t \mapsto t$ and $\psi \circ \varphi^{-1} : (\pi, 2\pi) \to (-\pi, 0)$ defined as $t \mapsto t - 2\pi$. Both these maps are smooth, and hence the charts are compatible. Thus, S^1 is a smooth manifold of dimension 1.

August 3rd.

Proposition 1.22. *Let $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ be an atlas on M. Suppose (V, ψ) and (W, σ) are two charts on M which are C^∞-compatible with all charts in $\mathcal{U}$. Then (V, ψ) and (W, σ) are C^∞-compatible.*

Proof. Let $p \in V \cap W$. Then there exists $\alpha \in \Lambda$ such that $p \in U_\alpha$. Since (V, ψ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\psi \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap V) \to \psi(U_\alpha \cap V)$ is smooth. Similarly, since (W, σ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\sigma \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap W) \to \sigma(U_\alpha \cap W)$ is smooth. Now, we have

$$(\sigma \circ \psi^{-1})|_{\psi(U_\alpha\cap V\cap W)} = (\sigma \circ \varphi_\alpha^{-1})|_{\varphi_\alpha(U_\alpha\cap V\cap W)} \circ (\varphi_\alpha \circ \psi^{-1})|_{\psi(U_\alpha\cap V\cap W)}. \tag{1.60}$$

The right-hand side is a composition of smooth maps, and hence is smooth. Thus, $\sigma \circ \psi^{-1}$ is smooth on an open neighbourhood of p. Since p was arbitrary, we have that $\sigma \circ \psi^{-1}$ is smooth on $V \cap W$. Similarly, we can show that $\psi \circ \sigma^{-1}$ is smooth on $V \cap W$. Thus, (V, ψ) and (W, σ) are compatible. ∎

This leads to the following definition.

Definition 1.23. A *maximal atlas* $\mathcal{M}$ on a smooth manifold M is an atlas which is maximal with respect to inclusion; that is, if $\mathcal{M} \subseteq \mathcal{U}$ for some atlas $\mathcal{U}$, then $\mathcal{M} = \mathcal{U}$. Equivalently, $\mathcal{M}$ is a maximal atlas if it contains all charts which are compatible with all charts in $\mathcal{M}$.

Any atlas $\mathcal{U}$ on M is contained in a unique maximal atlas $\mathcal{M}$. For a proof of this, let $\mathcal{M} = \{(U, \varphi) \mid$ compatible with all charts in $\mathcal{U}\}$. Then $\mathcal{M}$ is an atlas. If $\mathcal{M} \subseteq \mathcal{M}'$, then any chart in $\mathcal{M}'$ must be compatible with all charts in $\mathcal{M}$; it must be compatible with all charts in $\mathcal{U}$. By definition of $\mathcal{M}$, such a chart must be in $\mathcal{M}$, so $\mathcal{M} = \mathcal{M}'$. For uniqueness, if $\mathcal{U} \subseteq \mathcal{M}, \mathcal{M}'$, then $\mathcal{M}, \mathcal{M}' \subseteq \mathcal{M} \cup \mathcal{M}'$ tells us $\mathcal{M} = \mathcal{M}'$ by maximality. The unique maximal atlas $\mathcal{M} \supseteq \mathcal{U}$ is also called a differentiable structure on M.

Example 1.24.

1. Let $U \subseteq \mathbb{R}^n$, and $F : U \to \mathbb{R}^m$ be a C^∞-map. The graph of F is defined as $\Gamma(F) = \{(x, F(x)) \mid x \in U\} \subseteq \mathbb{R}^{n+m}$. At atlas on $\Gamma(F)$ is given by $\{(U, \varphi)\}$, where $\varphi : U \to \Gamma(F)$ is defined as $x \mapsto (x, F(x))$. The transition map is $\varphi^{-1} : \Gamma(F) \to U$ defined as $(x, F(x)) \mapsto x$. This is smooth, and hence $\Gamma(F)$ is a smooth manifold of dimension n.
2. Let M and N be smooth manifolds of dimension m and n, respectively. Then $M \times N$ is a smooth manifold of dimension $m + n$; if $\mathcal{U}$ and $\mathcal{V}$ are atlases on M and N, respectively, then $\mathcal{U} \times \mathcal{V} = \{(U \times V, \varphi \times \psi) \mid (U, \varphi) \in \mathcal{U}, (V, \psi) \in \mathcal{V}\}$ is an atlas on $M \times N$. The transition maps are $(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1} = (\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$, which is smooth since both $\varphi_2 \circ \varphi_1^{-1}$ and $\psi_2 \circ \psi_1^{-1}$ are smooth.

1.4.1 Smooth Functions and Smooth Maps

We will often omit 'smooth' but still mean smooth in the context of manifolds.

Definition 1.25. Let M be a manifold of dimension n. A function $f : M \to \mathbb{R}$ is smooth at $p \in M$ if there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth at $\varphi(p)$. A function $f : M \to \mathbb{R}$ is smooth if it is smooth at all $p \in M$.

The definition of the smooth function $f : M \to \mathbb{R}$ is independent of the choice of chart (U, φ), since if (V, ψ) is another chart containing p, then $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is smooth, and hence $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$ is smooth at $\psi(p)$.

Proposition 1.26. *Let M be a manifold of dimension n, and $f : M \to \mathbb{R}$ be a function. Then the following are equivalent:*

1. *f is a smooth function on M.*
2. *M has an atlas $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that $f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha) \to \mathbb{R}$ is smooth for all $\alpha \in \Lambda$.*
3. *For all charts (U, φ) on M, the map $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth.*

Proof. Note that 2. implies 1. by definition, and 3. implies 2. trivially. We show that 1. implies 3. Let (U, φ) be a chart on M, and $p \in U$; it is enough to prove that $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. Since f is smooth, there exists a chart (V, ψ) of M such that $p \in V$ and $f \circ \psi^{-1} : \psi(V) \to \mathbb{R}$ is smooth at $\psi(p)$. Now, we have

$$f \circ \varphi^{-1} = (f \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1}). \tag{1.61}$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. ∎

A function $f : M \to \mathbb{R}$ is smooth on a chart (U, φ) if and only if the pullback $\varphi^* f := f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth. Also note that the definition of a smooth function f does not assume that f is continuous; however, one may show that the definition itself implies that f is continuous. This is because if f is smooth at $p \in M$, then there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth at $\varphi(p)$, and hence continuous at $\varphi(p)$. Since φ is a homeomorphism, we have that f is continuous at p. Since p was arbitrary, we have that f is continuous on M.

Definition 1.27. Let M and N be smooth manifolds of dimension m and n, respectively. A continuous map $F : N \to M$ is smooth at $p \in N$ if there exist charts (U, φ) on N, with $p \in U$, and (V, ψ) on M, with $F(p) \in V$, such that $\psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is smooth at $\varphi(p)$. A continuous map $F : N \to M$ is smooth if it is smooth at all $p \in N$.

The definition of a map f being smooth at p is independent of the choice of charts (U, φ) and (V, ψ), since if (U', φ') and (V', ψ') are other charts containing p and $F(p)$, respectively, then $\psi' \circ \psi^{-1} : \psi(V \cap V') \to \psi'(V \cap V')$ and $\varphi' \circ \varphi^{-1} : \varphi(U \cap U') \to \varphi'(U \cap U')$ are smooth, and hence

$$\psi' \circ F \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1}) \tag{1.62}$$

Proposition 1.28. *Let $F : N \to M$ be a continuous map between smooth manifolds. Then the following are equivalent:*

1. *F is smooth.*
2. *There exist atlases $\mathcal{U}$ of N and $\mathcal{V}$ of M such that for all $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.*
3. *For all charts (U, φ) on N and (V, ψ) on M, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.*

Proof. Again, note that 2. implies 1. by definition, and 3. implies 2. trivially. A similar argument as in the previous proposition shows that 1. implies 3. ■

Proposition 1.29. *Let $F : N \to M$ and $G : M \to P$ be smooth maps between smooth manifolds. Then the composition $G \circ F : N \to P$ is smooth.*

Proof. Let (U, φ), (V, ψ), and (W, σ) be charts on N, M, and P, respectively, such that $p \in U$, $F(p) \in V$, and $G(F(p)) \in W$. Then we have

$$\sigma \circ (G \circ F) \circ \varphi^{-1} = (\sigma \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}). \tag{1.63}$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $G \circ F$ is smooth at p. ■

Finally, a smooth map $F : N \to M$ is termed a *diffeomorphism* if F is a bijection, and F and F^{-1} are smooth. Some properties of diffeomorphisms may be inferred.

Proposition 1.30. *The following hold true.*

1. *Let (U, φ) be a chart on a smooth manifold M of dimension n. Then $\varphi : U \to \varphi(U)$ is a diffeomorphism.*
2. *Suppose $U \subseteq M$ is open, and that $F : U \to F(U)$ is a diffeomorphism. Then (U, F) is a chart in the maximal atlas of M.*

Proof.

1. Note that U is a manifold since it is open in M, and $\varphi(U)$ is a manifold since it is open in $\mathbb{R}^n$. The map $\varphi : U \to \varphi(U)$ is a homeomorphism by definition of a chart, so it is bijective. Note that $\{(U, \varphi)\}$ is an atlas on U, and $\{(\varphi(U), \mathrm{id}_{\varphi(U)})\}$ is an atlas on $\varphi(U)$. Then φ is smooth since $\mathrm{id}_{\varphi(U)} \circ \varphi \circ \varphi^{-1} = \mathrm{id}_{\varphi(U)}$ is smooth, and φ^{-1} is smooth since $\varphi \circ \varphi^{-1} \circ \mathrm{id}_{\varphi(U)} = \mathrm{id}_{\varphi(U)}$ is smooth. Thus, φ is a diffeomorphism.

2. Since U is open in M, it is a smooth manifold. Because $F : U \to F(U)$ is a diffeomorphism, it is in particular a homeomorphism. Hence F is bijective, and its image $F(U)$ is an open subset of $\mathbb{R}^n$ (as required for a chart). Let (V, ψ) be any chart on M with $U \cap V \neq \varnothing$. Then the transition maps

$$\psi \circ F^{-1} : F(U \cap V) \to \psi(V) \quad \text{and} \quad F \circ \psi^{-1} : \psi(U \cap V) \to F(U \cap V) \tag{1.64}$$

are smooth, since they are compositions of smooth maps. Therefore (U, F) is compatible with every chart in the maximal atlas of M, and so (U, F) is a chart in that maximal atlas.

■

Proposition 1.31. *Let N be a smooth manifold of dimension n, and $F : N \to \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*
2. *N has an atlas $\mathcal{U}$ such that for all $(U, \varphi) \in \mathcal{U}$, the map $F \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^m$ is smooth.*
3. *For all charts (U, φ) on N, the map $F \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^m$ is smooth.*

Proof. 3. implies 2. is obvious. For 2. implies 1., note that there exist atlases $\mathcal{U}$ and $\mathcal{V}$ of N and M respectively such that $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$ satisfy the smoothness condition. Since $\mathbb{R}^m$ has a global chart $(\mathbb{R}^m, \mathrm{id}_{\mathbb{R}^m})$, we can take $\mathcal{V} = \{(\mathbb{R}^m, \mathrm{id}_{\mathbb{R}^m})\}$. Then for any $(U, \varphi) \in \mathcal{U}$, we have $\mathrm{id}_{\mathbb{R}^m} \circ F \circ \varphi^{-1} = F \circ \varphi^{-1}$ is smooth, which implies that F is smooth. For 1. implies 3., F is smooth and φ is a diffeomorphism; thus, φ^{-1} is smooth, and $F \circ \varphi^{-1}$ is smooth. ■

Proposition 1.32. *Let $F : N \to M$ be a continuous map between smooth manifolds. Then the following are equivalent.*

1. *F is smooth.*

2. *M has an atlas $\mathcal{V}$ such that for all $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F : F^{-1}(V) \to \psi(V)$ is smooth.*
3. *For any chart (V, ψ) of M, the map $\psi \circ F : F^{-1}(V) \to \psi(V)$ is smooth.*

The proof follows a similar structure to the one above.

Proposition 1.33. *Let $F : N \to \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*
2. *$F^1, \dots, F^m : N \to \mathbb{R}$ are smooth, where $F = (F^1, \dots, F^m)$.*

Examples

August 5th.

We look at some examples of smooth manifolds and smooth maps.

Example 1.34. For $S^n \subseteq \mathbb{R}^{n+1}$, the *$n$-sphere*, define $U_i^+ = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i > 0\}$ and $U_i^- = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i < 0\}$ for $i = 1, \dots, n+1$. Then $\{(U_i^+, \varphi_i^+), (U_i^-, \varphi_i^-)\}_{i=1}^{n+1}$ is an atlas on S^n, where $\varphi_i^+ : U_i^+ \to \mathbb{R}^n$ is defined as $(x_1, \dots, x_{n+1}) \mapsto (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1})$, and $\varphi_i^- : U_i^- \to \mathbb{R}^n$ is defined similarly. The transition maps are smooth, and hence S^n is a smooth manifold of dimension n.

Example 1.35. Recall $\mathbb{R}P^n$, the *real projective space* of dimension n, defined as the quotient $\mathbb{R}^{n+1} \setminus \{0\} / \sim$, where $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $(x_0, \dots, x_n) = (\lambda y_0, \dots, \lambda y_n)$. Let $(a_1 : a_2 : \dots : a_{n+1})$ denote the equivalence class of $(a_1, \dots, a_{n+1}) \in \mathbb{R}^{n+1} \setminus \{0\}$. This is a smooth manifold of dimension n; an atlas is given by $\{(U_i, \varphi_i)\}_{i=1}^{n+1}$, where $U_i = \{(x_1 : \dots : x_{n+1}) \in \mathbb{R}P^n \mid x_i \neq 0\}$ and $\varphi_i : U_i \to \mathbb{R}^n$ is defined as

$$\varphi_i(x_1 : \dots : x_{n+1}) = \left(\frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_{n+1}}{x_i} \right). \tag{1.65}$$

Verify that the transition maps $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ are smooth for all $i, j = 1, \dots, n+1$. We can also provide a smooth map $F : S^n \to \mathbb{R}P^n$ defined as $(x_1, \dots, x_{n+1}) \mapsto (x_1 : \dots : x_{n+1})$. This is a smooth map, and it is a 2-to-1 covering map.

Example 1.36. Similar to the above, we can define $\mathbb{P}_k^n$ to be the projective space of dimension n over the field k; that is, $\mathbb{P}_k^n = k^{n+1} \setminus \{0\} / \sim$, where $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in k \setminus \{0\}$ such that $(x_1, \dots, x_{n+1}) = (\lambda y_1, \dots, \lambda y_{n+1})$. In particular, if $k = \mathbb{C}$, then $\mathbb{P}_{\mathbb{C}}^n$ consists of complex 1-dimensional subspaces of $\mathbb{C}^{n+1}$. This is a smooth manifold of dimension $2n$. $\mathbb{R}P^n$ may be, thus, interpreted as the space of real 1-dimensional subspaces of $\mathbb{R}^{n+1}$. Building upon this, the *Grassmannian space* $\mathrm{Gr}(n, k)$ is defined as the space of k-dimensional subspaces of $\mathbb{R}^n$. This is a smooth manifold of dimension $k(n-k)$. The Grassmannian space $\mathrm{Gr}(n+1, 1)$ is precisely $\mathbb{R}P^n$, and $\mathrm{Gr}(n, 0)$ is a single point. $\mathrm{Gr}(n, n-1)$ is the space of hyperplanes in $\mathbb{R}^n$. Again, the Grassmannian construction may be generalized to any field.

1.4.2 Quotient Manifolds

August 10th.

Recall that for an equivalence relation $\sim$ on a set S, the quotient set $S/\sim$ is defined as the set of equivalence classes $[s] = \{t \in S \mid t \sim s\} \in S/\sim$. If S happens to be a topological space, then the projection map $\pi : S \to S/\sim$ defined as $s \mapsto [s]$ is continuous if we endow $S/\sim$ with the quotient topology; that is, a subset $U \subseteq S/\sim$ is open if and only if $\pi^{-1}(U) \subseteq S$ is open. This is the finest topology on $S/\sim$ which makes π continuous. Also recall the universal property; that is, if $f : S \to X$ is a set map between topological spaces, and $\pi : S \to S/\sim$ is the projection map, then there exists a unique map $\tilde{f} : S/\sim \to X$

such that $f = \tilde{f} \circ \pi$ if and only if f is constant on equivalence classes. In this case, $\tilde{f}$ is continuous if and only if f is continuous.

The equivalence relation $\sim$ on S is called open if the projection map $\pi : S \to S/\sim$ is open; that is, if $U \subseteq S$ is open, then $\pi(U) \subseteq S/\sim$ is open. Usually, $S/\sim$ is not T_2 is general. The following is a sufficient condition for $S/\sim$ to be T_2, or Hausdorff.

Proposition 1.37. *Let $\sim$ be an open equivalence relation on a topological space S. Then $S/\sim$ is T_2 if and only if $\{(x, y) \in S \times S \mid x \sim y\}$ is closed in $S \times S$.*

Proof. Suppose the set $R = \{(x, y) \in S \times S \mid x \sim y\}$ is closed in $S \times S$. That is, for all $(x, y) \in S \times S$ such that $x \not\sim y$, there exist open neighbourhoods U of x and V of y such that $U \times V \subseteq S \times S \setminus R$. Since $\sim$ is open, we have that $\pi(U)$ and $\pi(V)$ are open neighbourhoods of $[x]$ and $[y]$, respectively. We claim that $\pi(U) \cap \pi(V) = \emptyset$. Suppose not; then there exists $[z] \in \pi(U) \cap \pi(V)$. Then there exist $u \in U$ and $v \in V$ such that $[z] = [u] = [v]$, which implies that $u \sim v$. But this contradicts the fact that $(u, v) \in U \times V \subseteq S \times S \setminus R$. Thus, $\pi(U) \cap \pi(V) = \emptyset$, and hence $S/\sim$ is Hausdorff.

On the other hand, suppose $S/\sim$ is Hausdorff. Let $(x, y) \in S \times S$ such that $x \not\sim y$. Then there exist open neighbourhoods U of $[x]$ and V of $[y]$ in $S/\sim$ such that $U \cap V = \emptyset$. Since $\pi : S \to S/\sim$ is continuous, we have that $\pi^{-1}(U)$ and $\pi^{-1}(V)$ are open neighbourhoods of x and y, respectively. We claim that $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$. Suppose not; then there exists $(u, v) \in \pi^{-1}(U) \times \pi^{-1}(V)$ such that $u \sim v$. Then we have $[u] = [v]$, which implies that $[u] \in U \cap V$, contradicting the fact that $U \cap V = \emptyset$. Thus, $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$, and hence R is closed in $S \times S$. ∎

Example 1.38. Let $\mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\}$ be the punctured space, and define an equivalence relation $\sim$ on it as $(x_1, \ldots, x_{n+1}) \sim (y_1, \ldots, y_{n+1})$ if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $(x_1, \ldots, x_{n+1}) = (\lambda y_1, \ldots, \lambda y_{n+1})$. Then $\mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\}/\sim$ is precisely $\mathbb{R}P^n$. The projection map $\pi : \mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\} \to \mathbb{R}P^n$ is open; let $U \subseteq \mathbb{R}^n \setminus \{(0, \ldots, 0)\}$ be open. Then $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \mathbb{R} \setminus \{0\}} \lambda U$ is open, and hence $\pi(U)$ is open in $\mathbb{R}P^n$. Also, the set $R = \{(x, y) \in (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \mid x \sim y\}$ is closed in $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$, since if $y_i = \lambda x_i$ for some $\lambda \in \mathbb{R} \setminus \{0\}$, then $x_i y_j - x_j y_i = 0$ for all $i, j = 1, \ldots, n+1$. Thus, R is the zero set of the continuous functions $f_{ij} : (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \to \mathbb{R}$ defined as $(x, y) \mapsto x_i y_j - x_j y_i$ for all $i, j = 1, \ldots, n+1$. Hence, R is closed in $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$. By the previous proposition, we have that $\mathbb{R}P^n$ is Hausdorff.

Proposition 1.39. *Let $\sim$ be an open equivalence relation on a topological space S. Then S is second countable if and only if $S/\sim$ is second countable.*

Proof. Suppose S is second countable, and let $\mathcal{B}$ be a countable basis for S. Then $\mathcal{B}' = \{\pi(B) \mid B \in \mathcal{B}\}$ is a countable collection of open sets in $S/\sim$. We claim that $\mathcal{B}'$ is a basis for $S/\sim$. Let $U \subseteq S/\sim$ be open, and let $[x] \in U$. Then $\pi^{-1}(U) \subseteq S$ is open, and hence there exists $B \in \mathcal{B}$ such that $x \in B \subseteq \pi^{-1}(U)$. Then we have $[x] \in \pi(B) \subseteq U$, and hence $\mathcal{B}'$ is a basis for $S/\sim$. Thus, $S/\sim$ is second countable.

On the other hand, suppose $S/\sim$ is second countable, and let $\mathcal{B}'$ be a countable basis for $S/\sim$. Then $\mathcal{B} = \{\pi^{-1}(B') \mid B' \in \mathcal{B}'\}$ is a countable collection of open sets in S. We claim that $\mathcal{B}$ is a basis for S. Let $U \subseteq S$ be open, and let $x \in U$. Then $\pi(U) \subseteq S/\sim$ is open, and hence there exists $B' \in \mathcal{B}'$ such that $[x] \in B' \subseteq \pi(U)$. Then we have $x \in \pi^{-1}(B') \subseteq U$, and hence $\mathcal{B}$ is a basis for S. Thus, S is second countable. ∎

Example 1.40. Consider the disc $D^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_1^2 + \ldots + x_n^2 \leq 1\}$, and the hemisphere $H^n = \{(x_1, \ldots, x_{n+1}) \in S^n \mid x_{n+1} \geq 0\}$. On H^n, define an equivalence relation as $(x_1, \ldots, x_n, x_{n+1}) \sim (y_1, \ldots, y_n, y_{n+1})$ if either

1. $x_{n+1} = y_{n+1} = 0$ and $(y_1, \ldots, y_n, 0) = (-x_1, \ldots, -x_n, 0)$, or
2. $(x_1, \ldots, x_n, x_{n+1}) = (y_1, \ldots, y_n, y_{n+1})$.

On D^n, define an equivalence relation simply as $x \sim y$ if $x = y$ or $x = -y$. Then, one can show that $\mathbb{R}P^n \cong S^n/\sim \cong D^n/\sim \cong H^n/\sim$ as smooth manifolds, where $\sim$ on S^n is the antipodal equivalence relation.

1.5 Differential Structure

Let M be a smooth manifold of dimension n. For a point $p \in M$, let (U, φ) be a chart of M such that $p \in U$. The functions $r^1, \dots, r^n : \mathbb{R}^n \to \mathbb{R}$ defined as $r^i(x_1, \dots, x_n) = x_i$ are smooth, and hence, we can interpret (U, φ) as $(U, x^1, \dots, x^n)$, a chart with smooth coordinate functions $x^i = r^i \circ \varphi : U \to \mathbb{R}$ for $i = 1, \dots, n$. Then, given a map $f \in C^\infty(U)$, we define the partial derivative of f with respect to the coordinate function x^i at p as

$$\frac{\partial f}{\partial x^i}(p) := \frac{\partial (f \circ \varphi^{-1})}{\partial r^i}(\varphi(p)). \tag{1.66}$$

In fact, given any p in this particular U, we have

$$(\varphi^{-1})^* \left(\frac{\partial f}{\partial x^i} \right) = \frac{\partial f}{\partial x^i} \circ \varphi^{-1} = \frac{\partial (f \circ \varphi^{-1})}{\partial r^i} \in C^\infty(\varphi(U)). \tag{1.67}$$

This implies that $\frac{\partial f}{\partial x^i} \in C^\infty(U)$, and hence the partial derivative of f with respect to x^i is smooth on U. Moreover, one can show that $\frac{\partial x^i}{\partial x^j}(p) = \frac{\partial r^i}{\partial r^j}(\varphi(p)) = \delta^i_j$.

Let $F : N \to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let (U, φ) be a chart on N with $p \in U$, and (V, ψ) be a chart on M with $F(p) \in V$. Then we can interpret (U, φ) as $(U, x^1, \dots, x^n)$, and (V, ψ) as $(V, y^1, \dots, y^m)$, where $x^i = r^i \circ \varphi : U \to \mathbb{R}$ and $y^j = r^j \circ \psi : V \to \mathbb{R}$ are smooth coordinate functions. Then, we have $F^i = y^i \circ F = r^i \circ \psi \circ F \in C^\infty(U)$ for $i = 1, \dots, m$. The *Jacobian matrix* of F at p, with respect to the charts (U, φ) and (V, ψ), is defined as the $m \times n$ matrix

$$J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p) \right)_{1 \le i \le m, 1 \le j \le n}. \tag{1.68}$$

Each entry here can be written as

$$\frac{\partial F^i}{\partial x^j}(p) = \frac{\partial (\psi \circ F \circ \varphi^{-1})^i}{\partial r^j}(\varphi(p)) = \frac{\partial (r^i \psi F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial (y^i F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial (F^i \circ \varphi^{-1})}{\partial r^j}(\varphi(p)). \tag{1.69}$$

Recall the inverse function theorem, which states that if $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is smooth, and the Jacobian matrix $J_F(p)$ is invertible at $p \in U$, then there exists an open neighbourhood V of $F(p)$ such that $F : U \to V$ is a diffeomorphism. This motivates the following definition. Thus, if $m = n$ and $J_F(p)$ is invertible, then F is a local diffeomorphism at p; $\psi circ F \circ \varphi^{-1}$ is a local diffeomorphism at $\varphi(p)$ if and only if F is a local diffeomorphism at p.

Let M be a smooth manifold of dimension n, and let $p \in M$. Suppose $F^1, F^2, \dots, F^n$ are smooth functions on some open neighbourhood $U \subseteq M$ of p. Then consider $F = (F^1, \dots, F^n) : U \to \mathbb{R}^n$, which is a smooth map. Then $\left(\frac{\partial F^i}{\partial x^j}(p) \right)_{n \times n}$ is invertible if and only if F is a local diffeomorphism at p if and only if $F : W \to F(W) \subseteq \mathbb{R}^n$ is a diffeomorphism for some open neighbourhood $W \subseteq U$ of p. Then, on some open set around p, we can use $F^1, \dots, F^n$ as local coordinates. This motivates the following definition.

Definition 1.41. Let M be a smooth manifold, with $p \in M$. Recall $C^\infty_p(M) \equiv C^\infty_p$, the local ring of germs of smooth functions at p, which is also an $\mathbb{R}$-algebra. Also define T_pM to be the $\mathbb{R}$-vector space of derivations at p; that is, $T_pM = \{D : C^\infty_p \to \mathbb{R} \mid D \text{ is linear and satisfies the Leibniz rule}\}$. This T_pM is called the *tangent space* of M at p.

August 12th.

We leave it as an exercise to check that $\frac{\partial}{\partial x^i}|_p$ is a derivation from C^∞_p to $\mathbb{R}$, and hence $\frac{\partial}{\partial x^i}|_p \in T_pM$ for all valid i. Given a smooth map $F : N \to M$ between smooth manifolds, we can define the map $F_{*,p} : T_pN \to T_{F(p)}M$ as $D \mapsto F_{*,p}(D)$, where $F_{*,p}(D)(f) = D(f \circ F)$ for all $f \in C^\infty_{F(p)}$. This is the pushforward of f at p. Moreover, if $G : M \to P$ is another smooth map, then $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p}$. The map $F_{*,p}$ is linear, and hence a linear transformation between the tangent spaces T_pN and $T_{F(p)}M$.

If $F : N \to M$ were a diffeomorphism, then there exists a diffeomorphism $G : M \to N$ such that $G \circ F = \mathrm{id}_N$ and $F \circ G = \mathrm{id}_M$. Now look at the chain

$$T_pN \xrightarrow{F_{*,p}} T_{F(p)}M \xrightarrow{G_{*,F(p)}} T_{G(F(p))}N = T_pN. \tag{1.70}$$

Here, we have $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p} = (\mathrm{id}_N)_{*,p} = \mathrm{id}_{T_pN}$, and similarly $F_{*,p} \circ G_{*,F(p)} = \mathrm{id}_{T_{F(p)}M}$. Thus, $F_{*,p}$ is a linear isomorphism between the tangent spaces T_pN and $T_{F(p)}M$.

Let $p \in M$, a smooth manifold, and let (U, φ) be a chart containing p. Since $C^\infty_p(M) \equiv C^\infty_p(U)$, the diffeomorphism $\varphi : U \to \varphi(U)$ induces an isomorphism $\varphi_{*,p} : T_pM \to T_{\varphi(p)}\mathbb{R}^n$. We know that the tangent space on the right has the basis $\left\{\frac{\partial}{\partial r^1}|_{\varphi(p)}, \ldots, \frac{\partial}{\partial r^n}|_{\varphi(p)}\right\}$. Applying this map to a basis element, we have

$$\varphi_{*,p}\left(\frac{\partial}{\partial x^i}\Big|_p\right)(f) = \frac{\partial}{\partial x^i}\Big|_p (f \circ \varphi) = \frac{\partial}{\partial r^i}\Big|_{\varphi(p)} (f \circ \varphi \circ \varphi^{-1}) = \frac{\partial}{\partial r^i}\Big|_{\varphi(p)} (f) \quad \text{for all } f \in C^\infty_p(M). \tag{1.71}$$

Hence, $\varphi_{*,p}\left(\frac{\partial}{\partial x^i}|_p\right) = \frac{\partial}{\partial r^i}|_{\varphi(p)}$, and so $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$ is a basis for T_pM. Thus, we have shown that the tangent space T_pM has dimension n. This also shows that if $F : N \to M$ is a smooth map between smooth manifolds of dimension n and m that also happens to be a diffeomorphism, then $n = m$. Moreover, given a map $F : N \to M$ between smooth manifolds, and given a chart (U, φ) of $p \in N$ and a chart (V, ψ) of $F(p) \in M$, we have

$$F_{*,p} : T_pN \to T_{F(p)}M, \quad F_{*,p}\left(\frac{\partial}{\partial x^i}\Big|_p\right) = \sum_{j=1}^m b^i_j \frac{\partial}{\partial y^j}\Big|_{F(p)}, \quad b^i_j = \frac{\partial F^j}{\partial x^i}(p) \quad \text{for } i = 1, \ldots, n, j = 1, \ldots, m. \tag{1.72}$$

That is, the matrix of $F_{*,p}$ with respect to the bases $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$ and $\left\{\frac{\partial}{\partial y^1}|_{F(p)}, \ldots, \frac{\partial}{\partial y^m}|_{F(p)}\right\}$ is precisely the $m \times n$ real *Jacobian matrix* $(dF)_p := J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \le i \le m,\, 1 \le j \le n}$.

August 17th.

Suppose $0 \in (a, b) \subseteq \mathbb{R}$, and let $c : (a, b) \to M$ be a smooth map to a smooth manifold M of dimension n such that $c(0) = p \in M$. c is then called a *smooth curve* in M. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M with $p \in U$. Then $c^1(0) \in T_pM$ is defined as $c'(0) = c_{*,0}\left(\frac{d}{dt}\Big|_0\right)$, where $\frac{d}{dt}|_0 \in T_0\mathbb{R}$ is the standard basis element. Then $c'(0)$ is a tangent vector at p, and we have

$$\mathbb{R} \cong T_0((a, b)) \xrightarrow{c_{*,0}} T_pM \xrightarrow{\varphi_{*,p}} T_{\varphi(p)}\mathbb{R}^n \cong \mathbb{R}^n, \quad c'(0) \mapsto \varphi_{*,p}(c'(0)) = (\varphi \circ c)_{*,0}\left(\frac{d}{dt}\Big|_0\right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}\Big|_p \tag{1.73}$$

since $\{\frac{\partial}{\partial x^i}|_p\}_{i=1}^n$ is a basis for T_pM. Utilising $c'(0) = c_{*,0}\left(\frac{d}{dt}|_0\right)$, we have

$$a^i = \varphi_{*,p}(c'(0))(x^i) = c_{*,0}\left(\frac{d}{dt}\Big|_0\right)(x^i \circ \varphi^{-1}) = \frac{d}{dt}\Big|_0 (x^i \circ \varphi^{-1} \circ c) = \frac{d}{dt}\Big|_0 (x^i \circ c) = \frac{dc^i}{dt}(0). \tag{1.74}$$

Thus, a curve c can be thought of as a tangent vector at p, and the tangent vector is given by the derivative of the curve at 0. In particular, let us interpret $c : \mathbb{R} \to \mathbb{R}^n$ as a smooth curve in $\mathbb{R}^n$; we have $c(t) = (c_1(t), \ldots, c_n(t))$, and hence $c'(0) = (\frac{dc_1}{dt}(0), \ldots, \frac{dc_n}{dt}(0)) \in \mathbb{R}^n$. This is consistent with the above definition of $c'(0)$ as a tangent vector at p. From here, we wish to obtain a smooth map $c : (a, b) \to M$ such that $c(0) = p$ and $c'(0) = X_p = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}|_p$ for some $X_p \in T_pM$. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M with $p \in U$. In fact, choose a chart φ such that $\varphi(p) = 0 \in \mathbb{R}^n$ (translations are diffeomorphisms). Then we can define a smooth curve $c : (-\epsilon, \epsilon) \to M$ for some $\epsilon > 0$ such that $c(0) = p$ and $c'(0) = X_p$ as

$$c(t) = \varphi^{-1}(ta^1, \ldots, ta^n) \quad \text{for } t \in (-\epsilon, \epsilon). \tag{1.75}$$

Then, $c'(0) = c_{*,0}\left(\frac{d}{dt}|_0\right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}|_p = X_p$, as desired.

Coming back to the smooth map $F : N \to M$ between smooth manifolds, the rank of F at $p \in N$ is defined as the rank of the linear map $F_{*,p} : T_pN \to T_{F(p)}M$, which is also the rank of the matrix $(dF)_p = J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \le i \le m, 1 \le j \le n}$. The map $F : N \to M$ is called an *immersion* at $p \in N$ if $F_{*,p} : T_pN \to T_{F(p)}M$ is injective (that is, $\mathrm{rank}(F_{*,p}) = \dim(T_pN) = n$), and F is called an immersion if it is an immersion at all points $p \in N$. Similarly, F is called a *submersion* at $p \in N$ if $F_{*,p} : T_pN \to T_{F(p)}M$ is surjective (that is, $\mathrm{rank}(F_{*,p}) = \dim(T_{F(p)}M) = m$), and F is called a submersion if it is a submersion at all points $p \in N$. If $F : N \to M$ is a smooth map between smooth manifolds of dimension n and m, respectively, then $\mathrm{rank}(F_{*,p}) \le \min\{n, m\}$ for all $p \in N$. Thus, if $n < m$, then F cannot be a submersion, and if $n > m$, then F cannot be an immersion.

1.6 Regular Maps and Regular Submanifolds

Definition 1.42. Let $F : N \to M$ be a smooth map between smooth manifolds. $p \in N$ is called a *critical point* of F if $F_{*,p} : T_pN \to T_{F(p)}M$ is not surjective; that is, F is not a submersion at p. p is called a *regular point* of F if $F_{*,p} : T_pN \to T_{F(p)}M$ is surjective; that is, F is a submersion at p. Similarly, $q \in M$ is called a *critical value* of F if there exists a critical point $p \in N$ such that $F(p) = q$. $q \in M$ is called a *regular value* of F if it is not a critical value; that is, for all $p \in N$ such that $F(p) = q$, we have that p is a regular point of F. That is, either $F^{-1}(m) = \emptyset$, or $F_{*,p} : T_pN \to T_{F(p)}M$ is surjective for all $p \in F^{-1}(m)$.

From here, we can define something called a regular submanifold, which is useful in determining whether a certain set is a manifold or not. Let M be a manifold of dimension n, and $S \subseteq M$ be a subset with the subspace topology. Suppose that for all $p \in S$, there exists a chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ of M with $p \in U$ and $\varphi(p) = 0 \in \varphi(U) \subseteq \mathbb{R}^n$ such that

$$U \cap S = \{(x^1, x^2, \ldots, x^n) \in U \mid x^{i_1} = x^{i_2} = \ldots = x^{i_{n-k}} = 0 \text{ for some } 1 \leq i_1 < i_2 < \ldots < i_{n-k} \leq n\} \tag{1.76}$$

for some fixed integer $0 \leq k \leq n$ (via switching of coordinates, we may assume that the last $n - k$ coordinates are zero). Such a chart (U, φ) of M is said to be adapted to S. Then, we say that S is a *regular submanifold* of M of dimension k if for all $p \in S$, there exists a chart (U, φ) of M with $p \in U$ such that (U, φ) is adapted to S. Therefore, if (U, φ) is adapted to S, then $\varphi(U \cap S)$ consists of only those points that look like $(r^1, \ldots, r^k, 0, \ldots, 0) \in \mathbb{R}^n$, and hence $\varphi(U \cap S)$ is an open subset of $\mathbb{R}^k$. If we define $\varphi_S : U \cap S \to \mathbb{R}^k$ as $\varphi_S(p) = (x^1(p), \ldots, x^k(p))$, then $\varphi_S(U \cap S) = \varphi(U \cap S) \cap \mathbb{R}^k$.

Theorem 1.43. *Consider the maximal atlas $\mathcal{U} = \{(U, \varphi)\}$ of M, and let $\mathcal{A} = \{(U \cap S, \varphi_S) \mid (U, \varphi) \in \mathcal{U}\}$, where (U, φ) is adapted to S and $\bigcup\{U \cap S \mid (U, \varphi) \in \mathcal{U}\} = S$. Then $\mathcal{A}$ is a smooth atlas on S, and hence S is a smooth manifold of dimension k.*

The *codimension* of a regular submanifold S of M is defined as $\operatorname{codim}_M S = \dim(M) - \dim(S) = n - k$.

Let $f : M \to \mathbb{R}$ be a smooth function on a smooth manifold M of dimension n. Also let $0 \in \mathbb{R}$ be a regular value of f with $f^{-1}(0) \neq \emptyset$. Then, $f^{-1}(0)$ is a regular submanifold of M of dimension $n - 1$, or codimension 1. Note that 0 here is a red herring; we can replace 0 with any regular value $c \in \mathbb{R}$ of f with $f^{-1}(c) \neq \emptyset$, and the same result holds.

For a proof of this, let $p \in f^{-1}(0) \neq \emptyset$ so that $f(p) = 0$. Since 0 is a regular value of f, we have that $f_{*,p} : T_pM \to T_0\mathbb{R}$ is surjective. T_pM has a basis $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$, and $T_0\mathbb{R}$ has a basis $\left\{\frac{d}{dt}|_0\right\}$. The $1 \times n$ Jacobian matrix of f at p is given by

$$(df)_p = \left(\frac{\partial f}{\partial x^1}(p), \ldots, \frac{\partial f}{\partial x^n}(p)\right). \tag{1.77}$$

Since $f_{*,p} : T_pM \to T_0\mathbb{R}$ is surjective, we have that $\operatorname{rank}(f_{*,p}) = 1$, and hence $(df)_p \neq 0$; at least coordinate function x^i has $\frac{\partial f}{\partial x^i}(p) \neq 0$. Without the loss of generality, we may assume that $\frac{\partial f}{\partial x^1}(p) \neq 0$. Then, by the inverse function theorem, there exists an open neighbourhood V of p such that $f : V \to f(V) \subseteq \mathbb{R}$ is a diffeomorphism. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M with $p \in U \subseteq V$. Then, we have

$$U \cap f^{-1}(0) = \{(x^1, \ldots, x^n) \in U \mid x^1 = 0\}, \tag{1.78}$$

which shows that (U, φ) is adapted to $f^{-1}(0)$. Since $p \in f^{-1}(0)$ was arbitrary, we have that $f^{-1}(0)$ is a regular submanifold of M of dimension $n - 1$, or codimension 1. In general, this result generalizes to the following theorem.

Theorem 1.44. *Let $F : N \to M$ be a smooth map between smooth manifolds of dimension n and m, respectivel, with $n \geq m$. Let $q \in M$ be a regular value of F with $F^{-1}(q) \neq \emptyset$. Then, $F^{-1}(q)$ is a regular submanifold of N of dimension $n - m$, or codimension m.*

Proof. Let $p \in F^{-1}(q)$; then $F_{*,p} : T_pN \to T_qM$ is surjective. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart in N such that $\varphi(p) = 0 \in \mathbb{R}^n$, and let $(V, \psi) = (V, y^1, \ldots, y^m)$ be a chart in M such that $\psi(q) = 0 \in \mathbb{R}^m$.

We may assume $F(U) \subseteq V$. The matrix of $F_{*,p}$ with respect to the bases $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$ and $\left\{\frac{\partial}{\partial y^1}|_q, \ldots, \frac{\partial}{\partial y^m}|_q\right\}$ is given by $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1\leq i\leq m, 1\leq j\leq n}$; this has rank m since $F_{*,p}$ is surjective. Without the loss of generality, we may assume that the $m \times m$ submatrix $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1\leq i,j\leq m}$ is invertible. Then, by the inverse function theorem, there exists an open neighbourhood W of p such that $F : W \to F(W)$ is a diffeomorphism. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on N with $p \in U \subseteq W$. Then, we have

$$U \cap F^{-1}(q) = \{(x^1, \ldots, x^n) \in U \mid x^1 = x^2 = \ldots = x^m = 0\}, \tag{1.79}$$

which shows that (U, φ) is adapted to $F^{-1}(q)$. Since $p \in F^{-1}(q)$ was arbitrary, we have that $F^{-1}(q)$ is a regular submanifold of N of dimension $n - m$, or codimension m. ■

August 19th.

Example 1.45. Consider the map $f : \mathbb{R}^3 \to \mathbb{R}$ defined as $f(x, y, z) = x^2 + y^2 + z^2$. Then, $S^2 = f^{-1}(1) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$. We have $f_{*,p} : T_p\mathbb{R}^3 \to T_1\mathbb{R} = \mathbb{R}$ given by $f_{*,p}\left(\frac{\partial}{\partial x^i}|_p\right) = \sum_{j=1}^3 \frac{\partial f}{\partial x^j}(p)\frac{\partial}{\partial y}|_1$, where y is the coordinate function on $\mathbb{R}$. Then, the Jacobian matrix of f at p is given by $(df)_p = \left(\frac{\partial f}{\partial x}(p), \frac{\partial f}{\partial y}(p), \frac{\partial f}{\partial z}(p)\right) = (2x, 2y, 2z)$. The origin is the only critical point of f and is excluded from S^2. Thus, $1 \in \mathbb{R}$ is a regular value of f, and hence $S^2 = f^{-1}(1)$ is a regular submanifold of $\mathbb{R}^3$ of dimension $3 - 1 = 2$.

Example 1.46. Consider the determinant map $\det : \mathbb{R}^{n^2} \to \mathbb{R}$, interpreting a point $p \in \mathbb{R}^{n^2}$ as an $n \times n$ matrix $A \in M_n(\mathbb{R})$. Look at $SL_n(\mathbb{R}) = \det^{-1}(1) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}$. We have $\det_{*,A} : T_A\mathbb{R}^{n^2} \to T_1\mathbb{R} = \mathbb{R}$, where the bases are $\{\frac{\partial}{\partial x_{ij}}\}_{1\leq i,j\leq n}$ and $\{\frac{d}{dt}|_1\}$, respectively. Note that $\frac{\partial \det}{\partial x_{ij}}(A)$ is the (i, j)-th cofactor of A; not all of these cofactors can be zero since otherwise we would have $\det(A) = 0$ but $A \in \det^{-1}(1)$. Thus, $1 \in \mathbb{R}$ is a regular value of $\det$, and hence $SL_n(\mathbb{R}) = \det^{-1}(1)$ is a regular submanifold of $\mathbb{R}^{n^2}$ of dimension $n^2 - 1$.

The following is a corollary of the inverse function theorem; it may be generalized to the case of smooth maps between smooth manifolds.

Corollary 1.47. *Let $p \in \mathbb{R}^n$ have an open neighbourhood $U \subseteq \mathbb{R}^n$, and let $f : U \to \mathbb{R}^m$ be a smooth map with $f = (f_1, \ldots, f_m)$. Suppose f has rank k in a neighbourhood of p in U; that is, $\left(\frac{\partial f_i}{\partial x_j}(q)\right)$ has rank k for all q in this possibly smaller neighbourhood of p. Then there exists a diffeomorphism $G : V \to G(V)$, with $G(V)$ open in $\mathbb{R}^n$ and $p \in V$ open in U, and a diffeomorphism $F : W \to F(W)$, with $F(W)$ open in $\mathbb{R}^m$ and $f(p) \in W$ open in $\mathbb{R}^m$, such that $f(V) \subseteq W$, and $(FfG^{-1})(x^1, \ldots, x^n) = (x^1, \ldots, x^k, 0, \ldots, 0)$ for all $(x^1, \ldots, x^n) \in G(V)$.*

Proof. Let $p \in \mathbb{R}^n$ have an open neighbourhood $U \subseteq \mathbb{R}^n$, and let $f : U \to \mathbb{R}^m$ be a smooth map with $f = (f_1, \ldots, f_m)$. Suppose f has rank k in a neighbourhood of p in U. Then, without the loss of generality, we may assume that the $k \times k$ submatrix $\left(\frac{\partial f_i}{\partial x_j}(q)\right)_{1\leq i,j\leq k}$ is invertible for all q in this possibly smaller neighbourhood of p. Define the smooth map $F : U \to \mathbb{R}^n$ as $F(x^1, \ldots, x^n) = (f_1(x^1, \ldots, x^n), \ldots, f_k(x^1, \ldots, x^n), x^{k+1}, \ldots, x^n)$. Then, the Jacobian matrix of F at $q \in U$ is given by

$$(dF)_q = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(q) & \cdots & \frac{\partial f_1}{\partial x_k}(q) & \frac{\partial f_1}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_1}{\partial x_n}(q) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_k}{\partial x_1}(q) & \cdots & \frac{\partial f_k}{\partial x_k}(q) & \frac{\partial f_k}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_k}{\partial x_n}(q) \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.80}$$

Thus, by the inverse function theorem, there exists an open neighbourhood V of p with $V \subseteq U$ such that $F : V \to F(V)$ is a diffeomorphism. Consider the map $F^{-1} : F(V) \to V$; a point $(x^1, \ldots, x^n) \in F(V)$ is

mapped to $(x^1,\dots,x^k,f^{k+1},\dots,f^m)$. The Jacobian matrix of F^{-1} at $(x^1,\dots,x^n)\in F(V)$ is given by

$$(dF^{-1})_{(x^1,\dots,x^n)}=\begin{pmatrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0*&*&*&\frac{\partial f^{k+1}}{\partial x^{k+1}}&\dots&\frac{\partial f^{k+1}}{\partial x^n}*&*&*&\vdots&\ddots&\vdots*&*&*&\frac{\partial f^m}{\partial x^{k+1}}&\dots&\frac{\partial f^m}{\partial x^n}\end{pmatrix}. \tag{1.81}$$

We know this matrix to have rank k by hypothesis. Thus, the $(m-k)\times(n-k)$ submatrix $\left(\frac{\partial f^i}{\partial x^j}(q)\right)_{k+1\le i\le m,k+1\le j\le n}$ must have rank 0; that is, $\frac{\partial f^i}{\partial x^j}(q)=0$ for all $k+1\le i\le m$ and $k+1\le j\le n$. The functions $f^{k+1},\dots,f^m$ are then functions of only $x^1,\dots,x^k$. Define the smooth map $G:f(V)\to\mathbb{R}^m$ as $(y^1,\dots,y^m)=(y^1,\dots,y^k,y^{k+1}-f^{k+1},\dots,y^m-f^m)$. Then, the Jacobian matrix of G at $(y^1,\dots,y^m)\in f(V)$ is given by

$$(dG)_{(y^1,\dots,y^m)}=\begin{pmatrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0\\-\frac{\partial f^{k+1}}{\partial y^1}&\dots&-\frac{\partial f^{k+1}}{\partial y^k}&1&0&0\\\vdots&\ddots&\vdots&0&1&0\\-\frac{\partial f^m}{\partial y^1}&\dots&-\frac{\partial f^m}{\partial y^k}&0&0&1\end{pmatrix}. \tag{1.82}$$

This matrix is of full rank, so G is a diffeomorphism from $f(V)$ to $G(f(V))$. Then, we have $(GF)(x^1,\dots,x^n)=(f_1(x^1,\dots,x^n),\dots,f_k(x^1,\dots,x^n),0,\dots,0)$ for all $(x^1,\dots,x^n)\in V$. Thus, we have found diffeomorphisms G and F such that $(GF)(x^1,\dots,x^n)=(x^1,\dots,x^k,0,\dots,0)$ for all $(x^1,\dots,x^n)\in G(V)$. ■

The generalization is as follows. Let $f:N\to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let $p\in N$ be a point such that f has a rank k in a neighbourhood of p in N. Then there exists a chart (U,φ) centred at p in N and a chart (V,ψ) centred at $f(p)$ in M such that $f(U)\subseteq V$, and $\psi\circ f\circ\varphi^{-1}(x^1,\dots,x^n)=(x^1,\dots,x^k,0,\dots,0)$ for all $(x^1,\dots,x^n)\in\varphi(U)$.

August 24th.

Theorem 1.48. *Let $f:N\to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let $c\in M$ be such that f has a rank k in a neighbourhood of $f^{-1}(c)\neq\emptyset$ in N. Then $f^{-1}(c)$ is a regular submanifold of N of dimension $n-k$.*

Example 1.49. Look at the map $f:GL_n(\mathbb{R})\to GL_n(\mathbb{R})$ define by $f(A)=A^tA$. Then, $O_n(\mathbb{R})=f^{-1}(I_n)=\{A\in GL_n(\mathbb{R})\mid A^tA=I_n\}$. For now, the dimension of $O_n(\mathbb{R})$ will remain unknown, but we can still show that it is a regular submanifold of $GL_n(\mathbb{R})$. Let $C\in GL_n(\mathbb{R})$. Then $f(AC)=C^tf(A)C$. Note that conjugation by C is a diffeomorphism from $GL_n(\mathbb{R})$ to itself, and hence f has the same rank at A and AC. We shall show that the ranks of $f_{*,A}$ and $f_{*,B}$ are the same for all $A,B\in GL_n(\mathbb{R})$. We can break down $B=AC$ for some $C\in GL_n(\mathbb{R})$. Then $f\circ r_C=l_{C^t}\circ r_c\circ f$, where $r_C:GL_n(\mathbb{R})\to GL_n(\mathbb{R})$ is right multiplication by C and $l_{C^t}:GL_n(\mathbb{R})\to GL_n(\mathbb{R})$ is left multiplication by C^t. Then, we have

$$f_{*,AC}(r_C)_{*,A}=(l_{C^t})_{*,A^tAC}(r_C)_{*,A^tA}f_{*,A}. \tag{1.83}$$

The rank of $f_{*,A}$ is the same as the rank of $f_{*,AC}$ since $(r_C)_{*,A}$ and $(l_{C^t})_{*,A^tAC}(r_C)_{*,A^tA}$ are both isomorphisms. Thus, f has the same rank at all points in $GL_n(\mathbb{R})$.

The previous theorem may be generalized to the following theorem.

Theorem 1.50. *Let $f:N\to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Also let $S\subseteq M$ be a regular submanifold of codimension k, or dimension $m-k$. Suppose $f_{*,p}(T_pN)+T_{f(p)}S=T_{f(p)}M$ for all $p\in f^{-1}(S)$. Then, $f^{-1}(S)$ is a regular submanifold of N of*

codimension k, or dimension $n-k$.

Theorem 1.51. *Let $f : N \to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let $p \in N$.*

1. *Suppose* $\operatorname{rank}(f_{*,p}) = m$. *Then there exists a local chart (U, φ) centred at p in N and a local chart (V, ψ) centred at $f(p)$ in M such that $\psi f \varphi^{-1}(x^1, \ldots, x^n) = (x^1, \ldots, x^m)$ for all $(x^1, \ldots, x^n) \in \varphi(U)$.*

2. *Suppose* $\operatorname{rank}(f_{*,p}) = n$. *Then there exists a local chart (U, φ) centred at p in N and a local chart (V, ψ) centred at $f(p)$ in M such that $\psi f \varphi^{-1}(x^1, \ldots, x^n) = (x^1, \ldots, x^n, 0, \ldots, 0)$ for all $(x^1, \ldots, x^n) \in \varphi(U)$.*

Chapter 2

TANGENT SPACES, BUNDLES, AND VECTORS

2.1 Manifold Structure

August 31st.

Recall that given a smooth manifold M, the tangent space T_pM at a point $p \in M$ consists of all derivations from $C^\infty_p(M)$ to $\mathbb{R}$. T_pM exactly equals T_pU, where U is an open neighbourhood of p in M, since germs of functions need only match in a neighbourhood. If M is a manifold of dimension n, then any tangent space of a point in M is a vector space of dimension n, too. Now, the *tangent bundle* of the manifold is defined as the disjoint union of all tangent spaces:

$$TM = \bigsqcup_{p \in M} T_pM. \tag{2.1}$$

From here, let $\pi : TM \to M$ denote the projection map; that is, if $v \in T_pM \subseteq TM$ for some $p \in M$, then $\pi(v) = p$.

We can equip TM with a topology so that it becomes a smooth manifold is its own right, and so that π also happens to be a smooth map. For $p \in M$, let (U, φ) be a chart on M containing p. Note that for all $p \in U$, the tangent space T_pM has a basis $\{\frac{\partial}{\partial x^1}\big|_p, \dots, \frac{\partial}{\partial x^n}\big|_p\}$. Given any $v \in T_pM$, it can uniquely be written as

$$v = \sum_{i=1}^{n} c^i(v) \, \frac{\partial}{\partial x^i}\Big|_p. \tag{2.2}$$

Now, given $v \in TU$, we define a chart $(\tilde{U}, \tilde{\varphi})$ as $\tilde{U} = \pi^{-1}(U) = \bigsqcup_{p \in U} T_pM = TU$, and

$$\tilde{\varphi} : TU \to \varphi(U) \times \mathbb{R}^n, \quad v \mapsto (x^1(p), \dots, x^n(p), c^1(v), \dots, c^n(v)) = ((x^1 \circ \pi)(v), \dots, (x^n \circ \pi)(v), c^1(v), \dots, c^n(v)). \tag{2.3}$$

Note that $\tilde{\varphi}$ has a (set-theoretic) inverse since

$$\tilde{\varphi}^{-1} : \varphi(U) \times \mathbb{R}^n \to TU, \quad (\varphi(p), c^1, \dots, c^n) \mapsto v = \sum_{i=1}^{n} c^i \frac{\partial}{\partial x^i}\Big|_p. \tag{2.4}$$

On $\varphi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$, we put the subspace topology. Using the bijection $\tilde{\varphi}$, we put a topology on TU; that is, $A \subseteq TU$ is open if and only if $\tilde{\varphi}(A) \subseteq \varphi(U) \times \mathbb{R}^n$ is open.

With $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}$ is a maximal atlas of M, let $\mathcal{B} = \{A \subseteq TU_\alpha \mid A \text{ open and } \alpha \text{ is any index}\}$. We show that $\mathcal{B}$ is a basis for a topology on TM. The union being M is clear since

$$TM = \bigcup_{\alpha} TU_\alpha \subseteq \bigcup_{A \in \mathcal{B}} A \subseteq TM. \tag{2.5}$$

To show the second property of a basis, let $A \subseteq TU$ be open and $B \subseteq TV$ be open. Then $TU \cap TV = \pi^{-1}(U) \cap \pi^{-1}(V) = \pi^{-1}(U \cap V) = T(U \cap V)$ and so $A \cap B \subseteq T(U \cap V)$. Since $A \cap T(U \cap V)$ and $B \cap T(U \cap V)$ are both open in $T(U \cap V)$, $A \cap B = A \cap B \cap T(U \cap V)$ is open in $T(U \cap V)$.

Lemma 2.1. *TM is a second countable Hausdorff space.*

Proof. Let $v, w \in TM$; so that $v \in T_pM$ and $w \in T_qM$, or $\pi(v) = p$ and $\pi(w) = q$. Two cases arise. If $p \neq q$, then since M is Hausdorff, there exists open neighbourhoods $p \in U$ and $q \in V$ such that $U \cap V = \emptyset$. This simply implies $v \in \pi^{-1}(U) = TU$ and $w \in \pi^{-1}(V) = TV$, and $TU \cap TV = \emptyset$. For the case when $p = q$, both $v, w \in T_pM$. Since this T_pM identifies with $\mathbb{R}^n$, n being the dimension of M, we can pick open neighbourhoods $U_1, U_2 \subseteq T_pM$ of v and w, respectively, such that $U_1 \cap U_2 = \emptyset$. If we now pick a coordinate chart W of $p \in M$, then $W \times U_1$ and $W \times U_2$ are two open sets in TM that separate v and w, with $W \times U_1 \cap W \times U_2 = \emptyset$.

Let $\{B_i\}_{i \geq 1}$ be a countable basis of M, and let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ be an atlas. For $p \in U_\alpha$, there exists $p \in B_{p,\alpha} \subseteq U_\alpha$. Varying p and α, we get $\{B_{p,\alpha}\}_{p \in M,\, \alpha \in \Lambda} \subseteq \{B_i\}_{i \geq 1}$. So, there exists a countable basis of M consisting of coordinate charts for M. Since TU_i identifies with $\varphi_i(U_i) \times \mathbb{R}^n$ for any i, it is second countable and has a countable basis $\mathcal{B}_i$. Therefore, $\mathcal{B} = \bigcup_{i=1}^{\infty} \mathcal{B}_i$ is a countable basis of TM. ∎

Finally, we show that $\{(TU_\alpha, \tilde{\varphi}_\alpha)\}$ gives an atlas consisting of smooth compatible charts on TM. Let $(U, \varphi) = (U, x^1 \ldots, x^n)$ and $(V, \psi) = (V, y^1, \ldots, y^n)$ be two charts in M such that $U \cap V \neq \emptyset$. Let $p \in U \cap V$ with $v \in T_pM$. $\{\frac{\partial}{\partial x^1}\big|_p, \ldots, \frac{\partial}{\partial x^n}\big|_p\}$ and $\{\frac{\partial}{\partial y^1}\big|_p, \ldots, \frac{\partial}{\partial y^n}\big|_p\}$ are two bases for T_pM, so $v = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}\big|_p = \sum_{i=1}^n b^i \frac{\partial}{\partial y^i}\big|_p$. Applying y^i on both sides gives

$$b^i = \sum_{j=1}^{n} \frac{\partial y^i}{\partial x^j}(p) a^j. \tag{2.6}$$

This is a smooth transformation, showing that switching from one set of coordinates to another is smooth. Moreover, the TU_α's cover TM. This is a valid atlas.

As a consequence of this, the projection map $\pi : TM \to M$ is a smooth map, since $\varphi(U) \times \mathbb{R}^n \cong TU \xrightarrow{\pi} U \cong \varphi(U)$ takes the first n coordinates of $\varphi(U) \times \mathbb{R}^n$, which is a smooth map.

Definition 2.2. A *vector field* X on a smooth manifold M is a map $X : M \to TM$ such that $X(p) \in T_pM$ for all $p \in M$. In other words, a vector field is a smooth assignment of a tangent vector to each point in the manifold. It is termed smooth if the map X is smooth, meaning that for any chart (U, φ) on M, the composition $\tilde{\varphi} \circ X|_U : U \to \varphi(U) \times \mathbb{R}^n$ is smooth.

Such a vector field X is a section of the tangent bundle TM, as it assigns to each point $p \in M$ a vector in the corresponding tangent space T_pM. The projection map $\pi : TM \to M$ turns out to be the left inverse of X, since $\pi(X(p)) = p$ for all $p \in M$.

Definition 2.3. Let M be a smooth manifold of dimension n. A *smooth vector bundle* on M of rank r is a triple (E, M, π), where E is a smooth manifold of dimension $n + r$, and $\pi : E \to M$ is a smooth surjective map such that for each $p \in M$, the fiber $\pi^{-1}(p)$ is an r-dimensional real vector space. Additionally, π satisfies the local triviality condition: for every $p \in M$, there exists an open neighborhood U of p in M and a diffeomorphism $\phi_U : E_U := \pi^{-1}(U) \to U \times \mathbb{R}^r$ such that

$$\pi|_{E_U} = \mathrm{pr}_1 \circ \phi_U, \tag{2.7}$$

where $\mathrm{pr}_1 : U \times \mathbb{R}^r \to U$ is the projection onto the first factor, and $\varphi_U|_{\pi^{-1}(p)} : \pi^{-1}(p) \to \{p\} \times \mathbb{R}^r$ is a linear isomorphism of real vector spaces for each $p \in U$. Here, E is termed the *total space* and M the *base space* of the vector bundle.

Definition 2.4. A *morphism* between two smooth vector bundles (E, M, π) and (F, N, φ) is a pair of smooth maps $(\tilde{f}, f)$, where $\tilde{f} : E \to F$ and $f : M \to N$, such that $\varphi \tilde{f} = f\pi$, and for each $p \in M$, the map $\tilde{f}|_{\pi^{-1}(p)} : \pi^{-1}(p) \to \varphi^{-1}(f(p))$ is a linear map of real vector spaces.

As a special case, if $M = N$, and $f = \mathrm{id}_M$, then $\tilde{f}$ is called a *bundle map* over M. If $\tilde{f}$ is a diffeomorphism, then $(\tilde{f}, \mathrm{id}_M)$ is an isomorphism of vector bundles, and the two bundles are said to be isomorphic. In particular, if (E, M, π) is isomorphic to $(M \times \mathbb{R}^r, M, \mathrm{pr}_1)$, then (E, M, π) is called a *trivial vector bundle* of rank r.

2.1.1 Sections and Frames

September 2nd.

As earlier with a vector field, we have the following.

Definition 2.5. A map $s : M \to E$ is called a *section* of the vector bundle (E, M, π) if $\pi(s(p)) = p$ for all $p \in M$. In other words, a section assigns to each point in the base space a vector in the corresponding fiber. A section is termed smooth if the map s is smooth, meaning that for any chart (U, φ) on M, the composition $\phi_U \circ s|_U : U \to U \times \mathbb{R}^r$ is smooth.

We use $\Gamma(E) = \Gamma(M, E)$ to denote the set of all (global) smooth sections of the vector bundle (E, M, π). For any open subset $U \subseteq M$, we denote by $\Gamma(U, E)$ the set of all smooth sections of (E, M, π) over U; that is, smooth maps $s : U \to E$ such that $\pi(s(p)) = p$ for all $p \in U$. Moreover, for $s, t \in \Gamma(U, E)$, we define the sum $s + t \in \Gamma(U, E)$ by $(s + t)(p) = s(p) + t(p)$ for all $p \in U$; this sum is indeed in $\Gamma(U, E)$ since $\pi((s + t)(p)) = \pi(s(p) + t(p)) = \pi(s(p)) = p$, and $s + t$ can also be shown to be smooth. For $f \in C^\infty(U)$, we define the product $fs \in \Gamma(U, E)$ by $(fs)(p) = f(p)s(p)$ for all $p \in U$; this product is also in $\Gamma(U, E)$ since $\pi((fs)(p)) = \pi(f(p)s(p)) = \pi(s(p)) = p$, and fs can also be shown to be smooth. Therefore, $\Gamma(U, E)$ is a module over the ring $C^\infty(U)$, and $\Gamma(E)$ is a module over the ring $C^\infty(M)$.

Definition 2.6. A *frame* of a vector bundle (E, M, π) over an open subset $U \subseteq M$ is a collection of r sections $s_1, \ldots, s_r \in \Gamma(U, E)$ such that for each $p \in U$, the set $\{s_1(p), \ldots, s_r(p)\}$ forms a basis of the fiber $E_p = \pi^{-1}(p)$. The frame is termed smooth if each section s_i is smooth.

Consider the tangent bundle (TM, M, π) of a smooth manifold M of dimension n. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M. Then the collection of sections $s_i : U \to TM$ defined by $s_i(p) = \frac{\partial}{\partial x^i}\big|_p$ for $i = 1, \ldots, n$ forms a smooth frame of the tangent bundle over U.

Lemma 2.7. *Let (E, M, π) be a vector bundle, and let $U \subseteq M$ be an open subset. Suppose $s_1, \ldots, s_r$ is a smooth frame of E over U, and $s = \sum_{i=1}^r c^i s_i$. Then s is a smooth section of E over U if and only if each coefficient function $c^i : U \to \mathbb{R}$ is smooth.*

Proof. The "if" direction is clear from the definition of smoothness. For the "only if" direction, let s be a smooth section of E over U such that $s = \sum_{i=1}^r c^i s_i$. Let $p \in U$. We show that each c^i is smooth at p. Since $s_1, \ldots, s_r$ is a smooth frame of E over U, there exists an open neighborhood $V \subseteq U$ of p and a diffeomorphism $\phi_V : E_V \to V \times \mathbb{R}^r$ such that $\pi|_{E_V} = \mathrm{pr}_1 \circ \phi_V$, and $\phi_V|_{\pi^{-1}(q)} : \pi^{-1}(q) \to \{q\} \times \mathbb{R}^r$ is a linear isomorphism for each $q \in V$. Let $\phi_V(s(q)) = (q, (c^1(q), \ldots, c^r(q)))$ for all $q \in V$. Since $\phi_V \circ s|_V : V \to V \times \mathbb{R}^r$ is smooth, it follows that each component function $c^i : V \to \mathbb{R}$ is smooth. Therefore, each coefficient function c^i is smooth at p, and since p was arbitrary, each c^i is smooth on all of U. ∎

September 4th.

Definition 2.8. Let $f : N \to M$ be a smooth map between smooth manifolds, such that f is also injective and an immersion. Then $f(N)$, equipped with the topology borrowed from N, is termed an *immersed submanifold*.

In such a case, $f(N)$ is diffeomorphic to N. Note, however, that $f(N)$ need not be a manifold in the subspace topology of M; the subspace topology on $f(N)$ is weaker than the topology on $f(N)$ borrowed from N.

Definition 2.9. A smooth map $f : N \to M$ between smooth manifolds is called an *embedding* if f is injective and an immersion, and the subspace topology from M and the topology inherited from N are the same on $f(N)$.

Theorem 2.10. *If $S \subseteq M$ is a regular submanifold of M, then the inclusion map $i : S \to M$ is an embedding. Conversely, if $f : N \to M$ is an embedding, then $f(N)$ is a regular submanifold of M.*

Proof. Note that $i : S \to M$ is a smooth map, taking $(x^1, x^2, \ldots, x^s) \mapsto (x^1, x^2, \ldots, x^s, 0, \ldots, 0)$, is injective and an immersion. Also, the topologies trivially coincide. Hence, i is an embedding. For the converse, suppose $f : N \to M$ is an embedding. Let $p \in N$, with $f(p) \in f(N)$. Since f is an immersion (at p), there exist charts $(U, \varphi) = (U, x^1, \ldots, x^n)$, centred at p, and $(V, \psi) = (V, y^1, \ldots, y^m)$, centred at $f(p)$, such that $f(U) \subseteq V$, and f acts as $(x^1, \ldots, x^n) \to (x^1, \ldots, x^n, 0, \ldots, 0)$. Now, since $f(U) \subseteq f(N) \cap V$, we will change V to be the smallest open set $W \subseteq V \subseteq M$ such that $f(U) = f(N) \cap W$; since U is open in N if and only if $f(U)$ is open in $f(N)$ if and only if $f(U) = V' \cap f(N)$ where V' is open in M, we can set $W = V \cap V' \subseteq V$, so that $f(U) = W \cap f(N)$. ■

Theorem 2.11. *Let $f : N^n \to M^m$ be a smooth map between smooth manifolds. Suppose $S^s \subseteq M$ is a regular submanifold of M, and $f(N) \subseteq S$; this induces a set-theoretic factoring map $\tilde{f} : N \to S$ such that $i \circ \tilde{f} = f$. Then $\tilde{f}$ is a smooth map between smooth manifolds.*

Proof. With $p \in N$, and $f(p) \in S \subseteq M$, there exists a chart $(V, \psi) = (V, y^1, \ldots, y^m)$, centred at $f(p)$ in M, such that $S \cap V = \{(y^1, \ldots, y^m) \in V \mid y^{s+1} = \cdots = y^m = 0\}$. There also exits a chart $(U, \varphi) = (U, x^1, \ldots, x^n)$, centred at $p \in N$, such that $f(U) \subseteq V$ and f is given by $(x^1, \ldots, x^n) \mapsto (f^1, f^2, \ldots, f^m)$, where each f^i is smooth on U. But this means $f^{s+1}(x^1, \ldots, x^n) = \cdots = f^m(x^1, \ldots, x^n) = 0$ for all $(x^1, \ldots, x^n) \subseteq U$. In S, we have a smooth chart given by $(V \cap S, \varphi_S)$ via which, the map $\tilde{f}$ on U is given as $(x^1, \ldots, x^n) \mapsto (f^1, \ldots, f^s)$; $\tilde{f}$ is smooth. ■

2.2 Bump Functions and Partitions of Unity

September 18th.

Definition 2.12. A *bump function* is a smooth non-negative function f on a smooth manifold M such that f has compact support, that is, the closure of the set $\{q \in M \mid f(q) \neq 0\}$ is compact in M. A bump function is said to be adapted to an open set $U \subseteq M$ if the support of f is contained in U.

If $p \in U \subseteq M$ is an open set, then there exists a bump function f adapted to U such that $f(x) = 1$ in a neighbourhood of p contained in U.

Definition 2.13. A collection of subsets $\{A_\alpha\}_{\alpha \in \Lambda}$ of a topological space S is said to be *locally finite* if for all $p \in S$, there exists a neighbourhood V of P such that $\{\alpha \in \Lambda \mid A_\alpha \cap V \neq \emptyset\}$ is finite. In other words, each point in S has a neighbourhood that intersects only finitely many of the sets A_α.

Definition 2.14. A (smooth) *partition of unity* is a collection of smooth functions $\{\rho_\alpha : M \to \mathbb{R}\}_{\alpha \in \Lambda}$ on a smooth manifold M such that the following conditions hold:

1. $\{\operatorname{supp} \rho_\alpha\}_{\alpha \in \Lambda}$ is a locally finite collection of subsets of M.
2. $\sum_{\alpha \in \Lambda} \rho_\alpha(p) = 1$ for all $p \in M$.

Note that the sum in the second condition is well-defined since for each $p \in M$, there exists a neighbourhood V of p such that $\{\alpha \in \Lambda \mid \operatorname{supp} \rho_\alpha \cap V \neq \emptyset\}$ is finite, and so only finitely many terms in the sum are non-zero.

Theorem 2.15. *Let M be a smooth manifold, and let $\{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of M. Then the following hold true.*

1. *There exists a countable smooth partition of unity $\{\varphi_i\}_{i=1}^\infty$ on M such that for each i, $\operatorname{supp}(\varphi_i)$ is compact and contained in some U_α.*
2. *There exists a smooth partition of unity $\{\rho_\alpha\}_{\alpha\in\Lambda}$ on M such that for each α, $\operatorname{supp}(\rho_\alpha)$ is contained in U_α.*

In the second statement, we say that the partition of unity is *subordinate* to the open cover $\{U_\alpha\}_{\alpha\in\Lambda}$. Before we prove the theorem, we require a few results from topology.

Firstly, since M is a smooth manifold, it is second countable and has a countable basis $\{B_i\}_{i\geq 1}$. Define $\mathcal{S} = \{B_i \mid \overline{B_i} \text{ is compact}\}$. Then $\mathcal{S}$ is a countable basis of M consisting of open sets with compact closure. We shall show that $\mathcal{S}$ also happens to be a basis of M; to this end, let $p \in U \subseteq M$ be a point in an open set U. Moreover, let V be an open neighbourhood of p such that $\overline{V}$ is compact and $\overline{V} \subseteq U$. Since $\mathcal{B}$ is a basis of M, there exists $B_i \in \mathcal{B}$ such that $p \in B_i \subseteq V$. But then $\overline{B_i} \subseteq \overline{V}$ is compact, and so $B_i \in \mathcal{S}$. Therefore, $\mathcal{S}$ is a basis of M consisting of open sets with compact closure. Thus, any smooth manifold has a countable basis of open sets with compact closure.

Secondly, given any smooth manifold M, there exists a sequence of open sets $\{V_i\}_{i\geq 1}$ such that $\overline{V_i}$ is compact and $\overline{V_i} \subseteq V_{i+1}$ for all i, and $M = \bigcup_{i=1}^\infty V_i$. Indeed, let $\{B_i\}_{i=1}^\infty$ be a countable basis of M consisting of open sets with compact closure. Define $V_1 = B_1$, so that $V_1 \subseteq \overline{V}_1$ is compact. Thus, it must be contained in a finite union of the open cover $\{U_\alpha\}_{\alpha\in\Lambda}$; that is, there exists a finite set $A_1 \subseteq \Lambda$ such that $\overline{V}_1 \subseteq \bigcup_{\alpha\in A_1} U_\alpha$; for simplicity, suppose $\overline{V}_1 \subseteq B_1 \cup B_2 \cup \cdots \cup B_{i_1}$ for some $i_1 > 1$ such that i_1 is the smallest integer with this property. Now, let $V_2 = B_1 \cup B_2 \cup \cdots \cup B_{i_1}$, an open set with compact closure. Thus, $V_1 \subseteq \overline{V}_1 \subseteq V_2$, and $\overline{V}_2$ is compact. Repeating the same argument, we have $\overline{V}_2 \subseteq B_1 \cup B_2 \cup \cdots \cup B_{i_1} \cup \cdots \cup B_{i_2}$ for some $i_2 > i_1$ such that i_2 is the smallest integer with this property. Let $V_3 = B_1 \cup B_2 \cup \cdots \cup B_{i_2}$, an open set with compact closure. Thus, $V_2 \subseteq \overline{V}_2 \subseteq V_3$, and $\overline{V}_3$ is compact. Continuing in this manner, we get a sequence of open sets $\{V_i\}_{i=1}^\infty$ such that $\overline{V}_i$ is compact and $\overline{V}_i \subseteq V_{i+1}$ for all i, and $M = \bigcup_{i=1}^\infty V_i$. If we define $V_{-1} = V_0 = \emptyset$, we obtain a sequence as

$$V_{-1} \subseteq \overline{V}_{-1} \subseteq V_0 \subseteq \overline{V}_0 \subseteq V_1 \subseteq \overline{V}_1 \subseteq V_2 \subseteq \overline{V}_2 \subseteq V_3 \subseteq \cdots \tag{2.8}$$

From here, we can infer that

$$\overline{V}_{i+1} \setminus V_i \subseteq V_{i+2} \setminus \overline{V}_{i-1} \tag{2.9}$$

where $\overline{V}_{i+1} \setminus V_i$ is compact and $\overline{V}_{i-1} \subseteq V_{i+2}$ is open. We shall use ths construction in the proof of the theorem.

Proof. Start with the construction of the sequence of open sets $\{V_i\}_{i=-1}^\infty$ as above, where $V_{-1} = V_0 = \emptyset$, and $\overline{V}_i \subseteq V_{i+1}$ for all i. Let $p \in U_\alpha$ for some $\alpha \in \Lambda$. Then there exists i such that $p \in U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$. Since $U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$ is open, there exists a smooth bump function $\rho_p : M \to \mathbb{R}$ such that $\operatorname{supp}(\rho_p) \subseteq U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$ and $\rho_p > 0$ in some neighbourhood W_p of p. The collection $\{W_p\}_{p\in\overline{V}_{i+1}\setminus V_i}$ is then an open cover of the compact set $\overline{V}_{i+1} \setminus V_i$; refine this cover to a finite subcover consisting of $W_1, \ldots, W_m$, and let $\rho_1, \ldots, \rho_m$ be the corresponding bump functions. Repeating this for every i, we obtain, for each $\overline{V}_{i+1} \setminus V_i$, a finite cover $W_1^i, W_2^i, \ldots, W_{m(i)}^i$ of $\overline{V}_{i+1} \setminus V_i$, and corresponding bump functions $\rho_1^i, \rho_2^i, \ldots, \rho_{m(i)}^i$ such that $\operatorname{supp}(\rho_j^i) \subseteq U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$ for some $\alpha \in \Lambda$, and $\rho_j^i \equiv 1$ in W_j^i for all $j = 1, \ldots, m(i)$. Now, let $\{\rho_k\}_{k=1}^\infty$ be the countable collection of all these bump functions, and $\{W_j^i\}$ be the corresponding open sets. Note that the following properties hold:

1. $\operatorname{supp}(\rho_j^i)$ is compact for all i, j.
2. $\operatorname{supp}(\rho_j^i) \subseteq U_{\alpha_{ij}} \cap (V_{i+2} - \overline{V}_{i-1})$ for some $\alpha_{ij} \in \Lambda$.
3. $\overline{V}_{i+1} \setminus V_i \subseteq \bigcup_{j=1}^{m(i)} W_j^i$ for all i.
4. $\rho_j^i > 0$ on W_j^i for all i, j.

Since $\operatorname{supp}(\rho_j^i) \subseteq V_{i+2} \setminus \overline{V}_{i-1}$, we must have $\operatorname{supp}(\rho_j^i) \cap \overline{V}_{i-1} = \emptyset$ and $\operatorname{supp}(\rho_j^i) \cap (M \setminus V_{i+2}) = \emptyset$. Therefore, for each $p \in M$, there exists i such that $p \in V_{i+2} \setminus \overline{V}_{i-1}$, and so p can only belong to the support of bump

functions ρ_j^k for $k = i-1, i, i+1$. This shows that the collection $\{\text{supp}(\rho_j^i)\}$ is locally finite. Finally, we define

$$\varphi_k = \frac{\rho_k}{\sum_{j=1}^{\infty} \rho_j} \tag{2.10}$$

where the denominator is smooth and positive, and only finitely many terms are non-zero in a neighbourhood of any point. Then $\{\varphi_k\}_{k=1}^{\infty}$ is a countable smooth partition of unity on M such that for each k, $\text{supp}(\varphi_k)$ is compact and contained in some U_α. This proves the first statement. For the second statement, we can define $\rho_\alpha = \sum_{k \in I_\alpha} \varphi_k$, where $I_\alpha = \{k \mid \text{supp}(\varphi_k) \subseteq U_\alpha\}$. Then $\{\rho_\alpha\}_{\alpha \in \Lambda}$ is a smooth partition of unity on M subordinate to the open cover $\{U_\alpha\}_{\alpha \in \Lambda}$. ■

September 21st.

Lemma 2.16. *Let M be a smooth manifold, and X be a vector field on M. Then X is smooth on M if and only if for any chart (U, φ) on M, with $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$ on U, each coefficient function $a^i : U \to \mathbb{R}$ is smooth.*

Proof. Suppose X was smooth on M, or $X : M \to TM$ is a smooth map. Then for any chart (U, φ) on M, the composition $\tilde{\varphi} \circ X|_U : U \to \varphi(U) \times \mathbb{R}^n$ is smooth, and so each component function $a^i : U \to \mathbb{R}$ is smooth. Conversely, suppose that for any chart (U, φ) on M, with $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$ on U, each coefficient function $a^i : U \to \mathbb{R}$ is smooth. Then for any chart (U, φ) on M, the composition $\tilde{\varphi} \circ X|_U : U \to \varphi(U) \times \mathbb{R}^n$ is smooth, and so $X : M \to TM$ is a smooth map. ■

Proposition 2.17. *Let X be a vector field on a smooth manifold M. Then X is smooth if and only if for any smooth function $f \in C^\infty(M)$, the function $X(f) : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth.*

Proof. Suppose X is smooth. Then for any smooth function $f \in C^\infty(M)$, the function $X(f) : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. Conversely, suppose that for any smooth function $f \in C^\infty(M)$, the function $X(f) : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. Let (U, φ) be a chart on M, with $\varphi = (x^1, \ldots, x^n)$. Then on U, we have $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$, where each coefficient function $a^i : U \to \mathbb{R}$ is given by $a^i(p) = X_p(x^i)$. Since each coordinate function x^i is smooth, it follows that each coefficient function $a^i : U \to \mathbb{R}$ is smooth. Therefore, by the previous lemma, X is smooth on M. ■

Moreover, since X_p is a $\mathbb{R}$-linear derivation from $C^\infty(M)$ to $\mathbb{R}$, we get that X is a derivation on $C^\infty(M)$; that is, for any $f, g \in C^\infty(M)$, we have

$$X(fg)(p) = X_p(fg) = f(p)X_p(g) + g(p)X_p(f) = (fX(g) + gX(f))(p) \implies X(fg) = fX(g) + gX(f). \tag{2.11}$$

2.3 Integral Curves

The following theorem will be useful in the study of vector fields and their flows on smooth manifolds.

Theorem 2.18. *Let $p_0 \in V \subseteq \mathbb{R}^n$ be an open set, and $f : V \to \mathbb{R}^n$ be a smooth map. Then the differential equation $\frac{dy}{dt} = f(y)$ with initial condition $y(0) = p_0$ has a unique smooth solution $y : (a(p_0), b(p_0)) \to V$ where $(a(p_0), b(p_0))$ is the maximal open interval containing 0 such that y is defined on $(a(p_0), b(p_0))$.*

Definition 2.19. Let X be a smooth vector field on a smooth manifold M, and let $p \in M$. An *integral curve* of X is a smooth curve $c : (a, b) \to M$ such that $c'(t) = X_{c(t)}$ for all $t \in (a, b)$. We usually presume $0 \in (a, b)$; if $c(0) = p$, we say that c is an integral curve of X starting at p.
A *maximal integral curve* is an integral curve whose domain cannot be extended to a larger interval while still satisfying the integral curve condition.

We look at a few examples.

Example 2.20. Consider the smooth manifold $M = \mathbb{R}^2$ and the smooth vector field $X = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$. Let $p_0 = (1, 0) \in \mathbb{R}^2$. Then, an integral curve $c : (a, b) \to \mathbb{R}^2$ of X starting at p_0 satisfies the differential equation $c'(t) = (\dot{x}(t), \dot{y}(t)) = (-y(t), x(t)) = X_{c(t)}$ with initial condition $c(0) = (1, 0)$. Solving this system of differential equations, we get

$$c(t) = (\cos t, \sin t). \tag{2.12}$$

This integral curve is a circle centered at the origin with radius 1, and it is defined for all $t \in \mathbb{R}$. If, instead, we had chosen $p_0 = (x_0, y_0) \in \mathbb{R}^2$, then the integral curve of X starting at p_0 would be given by

$$c(t) = (x_0 \cos t - y_0 \sin t, x_0 \sin t + y_0 \cos t), \tag{2.13}$$

which is a circle centered at the origin with radius $\sqrt{x_0^2 + y_0^2}$, and it is also defined for all $t \in \mathbb{R}$. Moreover, one can verify that $c_s(c_t) = c_{s+t}$ for all $s, t \in \mathbb{R}$, where $c_t(p_0)$ denotes the integral curve of X starting at p_0 evaluated at time t. c_t is a diffeomorphism of $\mathbb{R}^2$ for each $t \in \mathbb{R}$, and the map

$$\mathbb{R} \to \mathrm{Diff}(\mathbb{R}^2), \quad t \mapsto c_t \tag{2.14}$$

is a smooth group homomorphism from $(\mathbb{R}, +)$ to the group of diffeomorphisms of $\mathbb{R}^2$, where the group operation on $\mathrm{Diff}(\mathbb{R}^2)$ is composition of maps.

From the above, any group homomorphism from $(\mathbb{R}, +)$ to the group of diffeomorphisms of a smooth manifold M is called a one-parameter group of diffeomorphisms of M.

Corollary 2.21. *Let X be a smooth vector field on a smooth manifold M, with $p \in M$. Then there exists an integral curve $c : (a, b) \to M$ of X such that $0 \in (a, b)$ with $c(0) = p$, and this integral curve is unique.*

Theorem 2.22. *Let $V \subseteq \mathbb{R}^n$ be open, and $f : V \to \mathbb{R}^n$ be a smooth map. Then for each point $p_0 \in V$, there exists a neighbourhood W of p_0 in V, an $\epsilon > 0$, and a smooth function $y : (-\epsilon, \epsilon) \times W \to V$ such that*

$$\frac{\partial y}{\partial t}(t, q) = f(y(t, q)), \qquad y(0, q) = q \tag{2.15}$$

for all $(t, q) \in (-\epsilon, \epsilon) \times W$.

Corollary 2.23. *Let X be a smooth vector field on a smooth manifold M, and let U be a chart around $p \in M$. Then there exists a neighbourhood W of p in U, an $\epsilon > 0$, and a smooth map $F : (-\epsilon, \epsilon) \times W \to U$ such that for all $q \in W$, $F(t, q)$ is an integral curve of X with initial point q ($F(0, q) = q$).*

In the above, we use the notation $F(t, q) = F_t(q)$. Suppose $s, t \in (-\epsilon, \epsilon)$, and $q \in W$, and that $F_t(F_s(q))$ and $F_{t+s}(q)$ are both defined. Then we have $F_t(F_s(q)) = F_{t+s}(q)$. This is because both sides are integral curves of X starting at q at time 0, and by uniqueness of integral curves, they must coincide. F, above, is termed as a *local flow* generated by the vector field X. $F_t(q)$ is then a flow line of q of the local flow F. If F can be defined as $F : \mathbb{R} \times M \to M$, then F is called a *global flow* generated by the vector field X. X is called a *complete vector field* if its flow is global.

Definition 2.24. In general, a *local flow* about a point $p \in U \subseteq M$ is a smooth map $F : (-\epsilon, \epsilon) \times W \to U$, where W is a neighbourhood of p in U and $\epsilon > 0$, such that

1. $F_0(q) = q$ for all $q \in W$, and
2. $F_t(F_s(q)) = F_{t+s}(q)$ whenever both sides are defined.

2.4 Lie Algebras and Lie Groups

September 23rd.

Recall that a vector field X on a smooth manifold M was smooth if and only if for any smooth function $f \in C^\infty(M)$, the function $Xf : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. That is, the map Xf itself is a smooth map on M; if $f \in C^\infty_p$, then $Xf \in C^\infty_p$ as well. If we have another smooth vector field Y on M, then we can define a new map $[X,Y](\cdot)(p) : C^\infty(M) \to \mathbb{R}$ by

$$f \mapsto X_p(Yf) - Y_p(Xf). \tag{2.16}$$

This map is $\mathbb{R}$-linear, and it is also a derivation on $C^\infty(M)$; that is, for any $f, g \in C^\infty(M)$, we have

$$\begin{aligned}
[X,Y](fg)(p) &= X_p(Y(fg)) - Y_p(X(fg)) = X_p(fYg + gYf) - Y_p(fXg + gXf) \\
&= f(p)X_p(Yg) + g(p)X_p(Yf) - f(p)Y_p(Xg) - g(p)Y_p(Xf) \\
&= f(p)(X_p(Yg) - Y_p(Xg)) + g(p)(X_p(Yf) - Y_p(Xf)) \\
&= f[X,Y](g)(p) + g[X,Y](f)(p).
\end{aligned} \tag{2.17}$$

Thus, $[X,Y](\cdot)(p)$ is a derivation on $C^\infty(M)$, and so there exists a unique tangent vector $[X,Y]_p \in T_pM$ such that $[X,Y]_p(f) = [X,Y](f)(p)$ for all $f \in C^\infty(M)$. This defines a new vector field $[X,Y]$ on M, defined as

$$XY - YX = [X,Y] : M \to TM, \quad p \mapsto [X,Y]_p. \tag{2.18}$$

This is called the Lie bracket of the vector fields X and Y, and is smooth when X and Y are smooth vector fields on M. $\mathfrak{X}(U)$ then, the set of all smooth vector fields on an open set $U \subseteq M$, is termed a Lie algebra.

Definition 2.25. Let k be a field and V be a vector space over k. V is termed a *Lie algebra* over k when equipped with an operation $[\cdot,\cdot] : V \times V \to V$ called the Lie bracket, such that the following conditions hold:

1. bilinearity: $[av_1 + bv_2, w] = a[v_1, w] + b[v_2, w]$ and $[v, aw_1 + bw_2] = a[v, w_1] + b[v, w_2]$ for all $v, v_1, v_2, w, w_1, w_2 \in V$ and $a, b \in k$,
2. anti-commutativity: $[v, w] = -[w, v]$ for all $v, w \in V$, and
3. Jacobi identity: $[v, [w, u]] + [w, [u, v]] + [u, [v, w]] = 0$ for all $v, w, u \in V$.

Verify that the set of smooth vector fields $\mathfrak{X}(U)$ on an open set $U \subseteq M$ forms a Lie algebra over $\mathbb{R}$ with the Lie bracket $[\cdot,\cdot] : \mathfrak{X}(U) \times \mathfrak{X}(U) \to \mathfrak{X}(U)$ defined above as $[X,Y] = XY - YX$ for all $X, Y \in \mathfrak{X}(U)$.

Any vector V over the field k can be viewed as a Lie algebra over k with the trivial Lie bracket $[v, w] = 0$ for all $v, w \in V$. This is also the only abelian Lie algebra over k; that is, a Lie algebra V over k is abelian if and only if $[v, w] = 0$ for all $v, w \in V$. The vector space $M_n(k)$ of all $n \times n$ matrices over a field k is a Lie algebra over k with the Lie bracket defined as $[A, B] = AB - BA$ for all $A, B \in M_n(k)$.

Definition 2.26. Let $F : N \to M$ be a smooth map between smooth manifolds, and let $p \in N$ with the induced linear map $F_{*,p} : T_pN \to T_{F(p)}M$. If X is a vector field on N and Y is a vector field on M, we say that X and Y are *F-related* if $F_{*,p}(X_p) = Y_{F(p)}$ for all $p \in N$. Y is called the pushforward of X by F.

Lemma 2.27. *With notation as above, X and Y are F-related if and only if for any smooth function $g \in C^\infty(M)$, we have $X(g \circ F) = (Yg) \circ F$.*

Proof. Suppose X and Y are F-related. Then for any smooth function $g \in C^\infty(M)$, we have

$$X(g \circ F) = F_{*,p}(X_p)(g) = Y_{F(p)}(g) = (Yg) \circ F.$$

Conversely, suppose that for any smooth function $g \in C^\infty(M)$, we have $X(g \circ F) = (Yg) \circ F$. Then for any smooth function $g \in C^\infty(M)$, we have

$$F_{*,p}(X_p)(g) = X(g \circ F) = (Yg) \circ F = Y_{F(p)}(g).$$

Thus, $F_{*,p}(X_p) = Y_{F(p)}$ for all $p \in N$, and so X and Y are F-related. ■

Lemma 2.28. *With notation as above, if X_1, X_2 are vector fields on N and Y_1, Y_2 are vector fields on M such that X_i and Y_i are F-related for $i = 1, 2$, then the Lie brackets $[X_1, X_2]$ and $[Y_1, Y_2]$ are also F-related.*

Proof. Suppose X_1, X_2 are vector fields on N and Y_1, Y_2 are vector fields on M such that X_i and Y_i are F-related for $i = 1, 2$. Then for any smooth function $g \in C^\infty(M)$, we have

$$\begin{aligned}
[X_1, X_2](g \circ F) &= X_1(X_2(g \circ F)) - X_2(X_1(g \circ F)) \\
&= X_1((Y_2 g) \circ F) - X_2((Y_1 g) \circ F) \\
&= (Y_1(Y_2 g)) \circ F - (Y_2(Y_1 g)) \circ F \\
&= ([Y_1, Y_2]g) \circ F.
\end{aligned}$$

Thus, by the previous lemma, we have that $[X_1, X_2]$ and $[Y_1, Y_2]$ are F-related. ■

September 25th.

Definition 2.29. A *Lie group* G is both a group and a smooth manifold such that the maps

$$G \times G \to G, \ \ (g, h) \mapsto gh, \qquad G \to G, \ \ g \mapsto g^{-1}. \tag{2.19}$$

are smooth maps.

For example, both $GL_n(\mathbb{R})$ and $SL_n(\mathbb{R})$ are Lie groups. Similarly, $O(n)$ is also a Lie group. $(\mathbb{R}^n, +)$ is also a Lie group. If G and H are both Lie groups, then so is $G \times H$.

Definition 2.30. If G and H are Lie groups, a *map of Lie groups* $f : G \to H$ is a map that is both a homomorphism of groups and a smooth map between smooth manifolds.

The determinant $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$ is a map of Lie groups.

Definition 2.31. Let G be a Lie group. A subset $H \subseteq G$ is termed a *Lie subgroup* if H is a subgroup and an immersed manifold of G, and the product and inverse operations restricted to H remain smooth.

Example 2.32. We look at the space $S^1 \times S^1 = \mathbb{R}^2/\mathbb{Z}^2$. Let f be the natural quotient map $f : \mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2$. Look at the map $g_\alpha : \mathbb{R} \to \mathbb{R}^2$ defined as $t \mapsto (t, \alpha t)$ for some $\alpha \in \mathbb{R}$. Let $L_\alpha = g_\alpha(\mathbb{R}) = \{(t, \alpha t) \mid t \in \mathbb{R}\}$ be the image line, and now set $H_\alpha = f(L_\alpha) = f \circ g_\alpha(\mathbb{R})$. We claim that $f \circ g_\alpha$ is a one-to-one immersion, and further that H_α is a closed curve if and only if α is rational. The map f may be explicitly written as $(s, t) \mapsto (e^{2\pi i s}, e^{2\pi i t})$. Note that it is enough to prove that $f^{-1}(H_\alpha) = \{(t + n, \alpha t + m) \mid t \in \mathbb{R}, n, m \in \mathbb{Z}\}$ is dense in $\mathbb{R}^2$ if and only if α is irrational. If $\alpha = \frac{p}{q}$ is rational, then $f^{-1}(H_\alpha) = \{(t + n, \frac{p}{q}t + m) \mid t \in \mathbb{R}, n, m \in \mathbb{Z}\}$. Then for any $t \in \mathbb{R}$, we have $(t+n, \frac{p}{q}t+m) = (t+n, \frac{p}{q}(t+qn)+(m-pn))$, and so $f^{-1}(H_\alpha) = \{(t+n, \frac{p}{q}t+m) \mid t \in [0, q), n, m \in \mathbb{Z}\}$. This is a union of q lines in $\mathbb{R}^2$, and so it cannot be dense in $\mathbb{R}^2$. Conversely, if α is irrational, then for any $(x, y) \in \mathbb{R}^2$ and $\epsilon > 0$, we can find integers n, m such that $|x - n| < \epsilon$ and $|y - m - \alpha n| < \epsilon$. This shows that $f^{-1}(H_\alpha)$ is dense in $\mathbb{R}^2$ if and only if α is irrational.

2.4.1 Matrix Lie Groups and Lie Algebras

September 28th.

Definition 2.33. For a matrix $A \in M_n(\mathbb{R})$, we define the exponential map $\exp : M_n(\mathbb{R}) \to GL_n(\mathbb{R})$ as

$$\exp(A) = e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \tag{2.20}$$

To show that the exponential map is well-defined and lands in $GL_n(\mathbb{R})$, we need to show that the series converges and that the resulting matrix is invertible. For $X \in M_n(\mathbb{R})$, define the norm $\|X\| = \sqrt{\sum_{i,j=1}^n x_{ij}^2}$; this norm is also sub-multiplicative, meaning that $\|XY\| \leq \|X\| \, \|Y\|$ for all $X, Y \in M_n(\mathbb{R})$. Then, looking at the partial sums of the series, we have

$$\left\| \sum_{k=0}^{N} \frac{A^k}{k!} \right\| \leq \sum_{k=0}^{N} \frac{\|A\|^k}{k!} \leq \|I\| + \sum_{k=1}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|} + \sqrt{n} - 1 \tag{2.21}$$

which implies that the series converges absolutely, and hence converges. We also state a few properties of the trace map $\operatorname{tr} : M_n(\mathbb{R}) \to \mathbb{R}$ defined as $\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}$ for $A = (a_{ij}) \in M_n(\mathbb{R})$; in particular, the following properties hold:

1. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for all $A, B \in M_n(\mathbb{R})$.
2. If $A \in GL_n(\mathbb{R})$, then $\operatorname{tr}(AXA^{-1}) = \operatorname{tr}(X)$ for all $X \in M_n(\mathbb{R})$.
3. For $A \in M_n(\mathbb{R})$, $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$, where $\lambda_1, \ldots, \lambda_n$ are the (complex) eigenvalues of A.
4. For all $A \in M_n(\mathbb{R})$, there exists $B \in GL_n(\mathbb{C})$ such that BAB^{-1} is upper triangular, and has the eigenvalues of A on the diagonal. In particular, $\det(A) = \prod_{i=1}^n \lambda_i$, where $\lambda_1, \ldots, \lambda_n$ are the (complex) eigenvalues of A.

Proposition 2.34. *For $A \in M_n(\mathbb{R})$, $\det e^A = e^{\operatorname{tr}(A)}$.*

Proof. Without the loss of generality, we may assume that A is upper triangular, with eigenvalues $\lambda_1, \ldots, \lambda_n$ on the diagonal (if not, we can find $B \in GL_n(\mathbb{C})$ such that BAB^{-1} is upper triangular, and $\det e^A = \det e^{BAB^{-1}}$ and $\operatorname{tr}(A) = \operatorname{tr}(BAB^{-1})$). Then e^A is also upper triangular, with eigenvalues $e^{\lambda_1}, \ldots, e^{\lambda_n}$ on the diagonal. Therefore, we have

$$\det e^A = \prod_{i=1}^{n} e^{\lambda_i} = e^{\sum_{i=1}^n \lambda_i} = e^{\operatorname{tr}(A)}. \tag{2.22}$$

■

In particular, if $A \in M_n(\mathbb{R})$, then $e^A \in GL_n(\mathbb{R})$, and so the exponential map $\exp : M_n(\mathbb{R}) \to GL_n(\mathbb{R})$ is well-defined. We use this exponential map with smooth curves. Define $c : \mathbb{R} \to GL_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$ as $c(t) = e^{tA}$ for some $A \in M_n(\mathbb{R})$. Then c is a smooth curve in $GL_n(\mathbb{R})$, with $c(0) = I$ and $c'(0) = A$. In fact, the derivative is given as

$$c'(t) = \frac{d}{dt} e^{tA} = \frac{d}{dt} \left(\sum_{k=0}^{\infty} \frac{t^k A^k}{k!} \right) = \sum_{k=1}^{\infty} \frac{k t^{k-1} A^k}{k!} = \sum_{k=1}^{\infty} \frac{t^{k-1} A^k}{(k-1)!} = A \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = A e^{tA}. \tag{2.23}$$

Now, let G be a Lie group, and let $g \in G$. Define the left translation map $\ell_g : G \to G$ as $\ell_g(h) = gh$ for all $h \in G$. Then ℓ_g is a diffeomorphism of G, with inverse $\ell_{g^{-1}}$. Let $G = GL_n(\mathbb{R})$. Then the induced linear map at the identity is

$$(\ell_g)_{*,I} : T_I(GL_n(\mathbb{R})) \to T_g(GL_n(\mathbb{R})) \tag{2.24}$$

With $c : \mathbb{R} \to GL_n(\mathbb{R})$ a smooth curve with $c(0) = I$ and $c'(0) = A$, we have

$$(\ell_g)_{*,I}(A) = (\ell_g \circ c)'(0) = (gc)'(0) = gc'(0) = gA. \tag{2.25}$$

The exact same map is induced. If we instead look at the map $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$, then the induced linear map at the identity is

$$(\det)_{*,I} : T_I(GL_n(\mathbb{R})) \to T_1(\mathbb{R}^*) \tag{2.26}$$

and repeating the above, we have

$$(\det)_{*,I}(A) = (\det \circ c)'(0) = \frac{d}{dt}\det(c(t))\bigg|_{t=0} = \frac{d}{dt}\det(e^{tA})\bigg|_{t=0} = \frac{d}{dt}e^{\mathrm{tr}(tA)}\bigg|_{t=0} = \mathrm{tr}(A). \tag{2.27}$$

That is, the induced linear map $(\det)_{*,I} : T_I(GL_n(\mathbb{R})) \to T_1(\mathbb{R}^*)$ is given as $A \mapsto \mathrm{tr}(A)$. Now, pre-add the inclusion map $i : SL_n(\mathbb{R}) \to GL_n(\mathbb{R})$, and consider the composition $\det \circ i : SL_n(\mathbb{R}) \to \mathbb{R}^*$. Then the induced linear map at the identity is

$$(\det \circ i)_{*,I} : T_I(SL_n(\mathbb{R})) \to T_1(\mathbb{R}^*). \tag{2.28}$$

But with the chain rule for induced linear maps, we have

$$(\det \circ i)_{*,I} = (\det)_{*,I} \circ (i)_{*,I} : T_I(SL_n(\mathbb{R})) \to T_1(\mathbb{R}^*), \qquad A \mapsto \mathrm{tr}(A) = 0. \tag{2.29}$$

That is, the induced linear map $(\det \circ i)_{*,I} : T_I(SL_n(\mathbb{R})) \to T_1(\mathbb{R}^*)$ is the zero map. This shows that $T_I(SL_n(\mathbb{R})) = \ker((\det \circ i)_{*,I}) = \{A \in M_n(\mathbb{R}) \mid \mathrm{tr}(A) = 0\}$, which is a vector space of dimension $n^2 - 1$. From here, we define

$$\mathfrak{sl}_n(\mathbb{R}) := T_I(SL_n(\mathbb{R})) = \{A \in M_n(\mathbb{R}) \mid \mathrm{tr}(A) = 0\}, \tag{2.30}$$

the space of traceless matrices. Moving on, we look at the Lie algebra of $O(n)$. Let $G = O(n) \subseteq GL_n(\mathbb{R})$ and consider the map $f : GL_n(\mathbb{R}) \to GL_n(\mathbb{R})$ defined as $f(A) = A^T A$. Then $O(n) = f^{-1}(I)$, and so the induced linear map at the identity is

$$f_{*,I} : T_I(GL_n(\mathbb{R})) \to T_I(GL_n(\mathbb{R})). \tag{2.31}$$

Pre-composing the smooth curve $c : \mathbb{R} \to GL_n(\mathbb{R})$ defined as $c(t) = e^{tA}$, we have

$$f_{*,I}(A) = (f \circ c)'(0) = (f(e^{tA}))'(0) = (e^{tA^T}e^{tA})'(0) = (e^{t(A^T+A)})'(0). \tag{2.32}$$

Thus, we have $T_I(O(n)) = \ker(f_{*,I}) = \{A \in M_n(\mathbb{R}) \mid A^T + A = 0\}$, which is a vector space of dimension $\frac{n(n-1)}{2}$. From here, we define

$$\mathfrak{so}_n(\mathbb{R}) := T_I(O(n)) = \{A \in M_n(\mathbb{R}) \mid A^T + A = 0\}, \tag{2.33}$$

the space of skew-symmetric matrices. We also denote $\mathfrak{gl}_n(\mathbb{R}) := T_I(GL_n(\mathbb{R})) = M_n(\mathbb{R})$, the space of all $n \times n$ matrices. These structures are all Lie algebras (subalgebras of $\mathfrak{gl}_n(\mathbb{R})$) with the Lie bracket defined as $[A, B] = AB - BA$ for all $A, B \in \mathfrak{gl}_n(\mathbb{R})$. In particular, we have the inclusions

$$\mathfrak{so}_n(\mathbb{R}) \subseteq \mathfrak{sl}_n(\mathbb{R}) \subseteq \mathfrak{gl}_n(\mathbb{R}). \tag{2.34}$$

It is left as an exercise to show that the Lie bracket is closed on $\mathfrak{so}_n(\mathbb{R})$ and $\mathfrak{sl}_n(\mathbb{R})$.

One may show that if $AB = BA$, then $e^{A+B} = e^A e^B$, and $(e^A)^{-1} = e^{-A}$.

Definition 2.35. Let G be a Lie group, and X be a (not necessarily smooth) vector field on G. X, then, is called a *left-invariant vector field* on G if $(\ell_g)_* X = X$ for all $g \in G$, or

$$(\ell_g)_* X_h = X_{\ell_g(h)} = X_{gh} \tag{2.35}$$

for any $g, h \in G$. That is, X is left-invariant if it is invariant under the pushforward of left translation by any element of G.

If we pick the identity $e \in G$, the above property of the vector field X boils down to the property that $X_g = (\ell_g)_* X_e$ for all $g \in G$. That is, a left-invariant vector field is completely determined by its value at the identity.

Proposition 2.36. *Any left-invariant vector field on a Lie group G is smooth.*

Proof. Let X be a left-invariant vector field on a Lie group G. Note that it is enough to prove that if $f \in C^\infty(G)$, then $Xf \in C^\infty(G)$. Let $g \in G$, and let $c : \mathbb{R} \to G$ be a smooth curve with $c(0) = e$ and $c'(0) = X_e$. Then, we have

$$(Xf)(g) = X_g(f) = (\ell_g)_* X_e(f) = X_e(f \circ \ell_g) = \frac{d}{dt}(f \circ \ell_g \circ c(t))\bigg|_{t=0}. \tag{2.36}$$

The map $(g, t) \mapsto f \circ \ell_g \circ c(t)$ is smooth, and so the map $g \mapsto (Xf)(g)$ is smooth as well. Thus, $Xf \in C^\infty(G)$, and so X is a smooth vector field on G. ∎

Suppose X and Y are two left-invariant vector fields on a Lie group G. That means, X and X, and Y and Y are ℓ_g-related for all $g \in G$. Thus, by the result above, $[X, Y]$ and $[X, Y]$ must be ℓ_g-related, which tells us $[X, Y]$ is a left-invariant vector field on G. Thus, we have, from before,

$$\mathfrak{X}(G) = \{X : G \to TG \mid X \text{ is smooth}\}, \tag{2.37}$$

the set of all smooth vector fields on G along with the Lie bracket $[X, Y] = XY - YX$ is a Lie algebra, and

$$L(G) := \{X : G \to TG \mid X \text{ is left-invariant}\} \subseteq \mathfrak{X}(G) \tag{2.38}$$

is a Lie subalgebra of $\mathfrak{X}(G)$. That is, there is a bijection between left-invariant vector fields on G and the tangent space at the identity T_eG. In particular, we have

$$L(G) \cong T_eG, \qquad X \mapsto X_e \tag{2.39}$$

where the inverse map is given as $v \mapsto X_v$, where X_v is the unique left-invariant vector field on G such that $X_{v,e} = v$. Thus, we have that $L(G)$ is a finite-dimensional Lie algebra over $\mathbb{R}$, with dimension $\dim L(G) = \dim T_eG = \dim G$. This bijection is as $\mathbb{R}$-linear spaces, and the Lie bracket on $L(G)$ is given as $[X_v, X_w] = X_{[v,w]}$ for all $v, w \in T_eG$. The Lie algebra $L(G)$ is called the Lie algebra of the Lie group G. In particular, we have that $L(GL_n(\mathbb{R})) = \mathfrak{gl}_n(\mathbb{R})$, $L(SL_n(\mathbb{R})) = \mathfrak{sl}_n(\mathbb{R})$, and $L(O(n)) = \mathfrak{so}_n(\mathbb{R})$.

September 30th.

We state a few facts, summarising the above discussion.

1. Any left-invariant vector field X on a Lie group G is smooth.
2. If X annd Y are two left-invariant vector fields on G, then $[X, Y]$ is also a left-invariance vector field.
3. $L(G)$, the set of all (smooth) left-invariant vector fields on G, is a Lie subalgebra of $\mathfrak{X}(G)$, the Lie algebra of all smooth vector fields on G.
4. There is an $\mathbb{R}$-linear isomorphism $T_eG \cong L(G)$; specifically, the map $X \mapsto X_e$ for $X \in L(G)$ and the map $A \mapsto \tilde{A}$ for $A \in T_eG$, define as $(\tilde{A})_h = (\ell_h)_* A$ for all $h \in G$, are inverses of each other.
5. T_eG becomes a Lie algebra via the above isomorphism, with the Lie bracket defined as $[A, B] = [\tilde{A}, \tilde{B}]_e$ for all $A, B \in T_eG$.

For $F : G \to H$ is a homomorphism of Lie groups, define $F_* : L(G) \to L(H)$ as $F_*X = \widetilde{F_{*,e}(X_e)}$ for all $X \in L(G)$. Then F_* is a homomorphism of Lie algebras. F_* is termed the pushforward of a left-invariant vector field X on G by the homomorphism F.

Proposition 2.37. *Let $F : G \to H$ be a homomorphism of Lie groups. Let $X \in L(G)$ and $F_*X \in L(H)$ be the pushforward of X by F. Then X and F_*X are F-related.*

Proof. We are to show that $F_{*,g}(X_g) = (F_*X)_{F(g)}$ for all $g \in G$. We have, via $X_g = (\ell_g)_{*,e}(X_e)$,

$$\begin{aligned} F_{*,g}((\ell_g)_{*,e}(X_e)) &= (F \circ \ell_g)_{*,e}(X_e) = (\ell_{F(g)} \circ F)_{*,e}(X_e) \\ &= (\ell_{F(g)})_{*,F(e)}(F_{*,e}(X_e)) = (\ell_{F(g)})_{*,F(e)}((F_*X)_{F(e)}) = (F_*X)_{F(g)}. \end{aligned} \tag{2.40}$$

∎

Proposition 2.38. *Let $F : G \to H$ be a homormophism of Lie groups, and set $\mathfrak{g} = T_eG$ and $\mathfrak{h} = T_eH$, their associated Lie algebras. Then the induced linear map $F_{*,e} : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism of Lie algebras.*

Proof. Let $A, B \in \mathfrak{g}$. Since $\tilde{A}$ and $F_*\tilde{A}$, and $\tilde{B}$ and $F_*\tilde{B}$ are F-related, we have that $[\tilde{A}, \tilde{B}]$ and $[F_*\tilde{A}, F_*\tilde{B}]$ are also F-related. Thus, we have

$$F_{*,e}([A, B]) = F_{*,e}([\tilde{A}, \tilde{B}]_e) = [F_*\tilde{A}, F_*\tilde{B}]_{F(e)} = [F_{*,e}(A), F_{*,e}(B)]. \tag{2.41}$$

■

Theorem 2.39. *Let G be a Lie group, and $\mathfrak{g} = T_eG$ be its associated Lie algebra. Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a Lie subalgebra. Then there exists a unique connected Lie subgroup $H \subseteq G$ such that $T_eH = \mathfrak{h}$.*

Corollary 2.40. *There exists a bijective correspondence between connected Lie subgroups of a Lie group G and Lie subalgebras of its associated Lie algebra $\mathfrak{g} = T_eG$.*

Theorem 2.41. *Let H be a Lie subgroup of a Lie group G. Then H is a closed subgroup of G if and only if H is a regular submanifold of G.*

Theorem 2.42. *Let G and H be Lie groups with associated Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Suppose G is simply connected. Then, given a Lie algebra homomorphism $\psi : \mathfrak{g} \to \mathfrak{h}$, there exists a unique homomorphism of Lie groups $\varphi : G \to H$ such that $\varphi_{*,e} = \psi$.*

Corollary 2.43. *If G and H are simply connected Lie groups with isomorphic associated Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, then G and H are isomorphic as Lie groups.*

Corollary 2.44. *There exits a bijective correspondence between isomorphism classes of Lie algebras and isomorphism classes of simply connected Lie groups.*

Theorem 2.45. *Let G and H be Lie groups. Then any continuous Lie group homomorphism $\varphi : G \to H$ is smooth.*

Corollary 2.46. *A second countable Hausdorff locally Euclidean topological group G has at most one differentiable structure making it into a Lie group.*

Theorem 2.47. *Let G be a lie group, and H be a closed (not necessarily Lie) subgroup of G. Then H has a unique manifold structure making it into a Lie subgroup of G.*

Theorem 2.48. *Let $\psi : G \to H$ be a homomorphism of Lie groups, and set $A = \ker \psi$ and $\mathfrak{A} = \ker \psi_{*,e}$, where $\psi_{*,e} : \mathfrak{g} \to \mathfrak{h}$ is the induced linear map between the associated Lie algebras. Then A is a closed Lie subgroup of G whose associated Lie algebra is $\mathfrak{A}$.*

Chapter 3

FORMS, ORIENTATION, AND INTEGRATION

3.1 Differential Forms

October 5th.

We translate our theory of differential forms on $\mathbb{R}^n$ onto smooth manifolds. A 1-form on a manifold M is section of the cotangent bundle T^*M. That is, a 1-form ω on M is a map $\omega : M \to T^*M = \bigsqcup_{p\in M} T_p^*M$ such that $\omega(p) \in T_p^*M$ for all $p \in M$. This is analagous to a vector field X which is a map $X : M \to TM = \bigsqcup_{p\in M} T_pM$ such that $X(p) \in T_pM$ for all $p \in M$. Given a smooth function $f : M \to \mathbb{R}$, we have the differential $df : M \to T^*M$ defined as $df(p) = df_p$, where $df_p : T_pM \to \mathbb{R}$ is the linear map defined as $df_p(X_p) = X_p(f)$ for all $X_p \in T_pM$. In particular, we have that df is a 1-form on M.

Moreover, given a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M, the x^i's can, again, be thought of as coordinate functions. With $\{\frac{\partial}{\partial x^i}\}_{i=1}^n$ the standard basis for T_pM, p being in U, the partials $\{(dx^i)_p\}$ forms the respective dual basis for T_p^*M. Thus, any $\omega_p \in T_p^*M$ can be written as

$$\omega_p = \sum_{i=1}^{n} a_i(p)(dx^i)_p \tag{3.1}$$

where $a_i(p)$'s are real numbers. More generally, any 1-form ω on U can be written as

$$\omega = \sum_{i=1}^{n} a_i dx^i \tag{3.2}$$

where $a_i : U \to \mathbb{R}$ are real-valued functions on U.

Definition 3.1. The *cotangent bundle* of a smooth manifold M is the vector bundle $T^*M = \bigsqcup_{p\in M} T_p^*M$ over M, with projection $\pi : T^*M \to M$ defined as $\pi(\omega_p) = p$ for all $\omega_p \in T_p^*M$. A *1-form* on M is a section of the cotangent bundle, that is, a map $\omega : M \to T^*M$ such that $\omega(p) \in T_p^*M$ for all $p \in M$.

We can, again, put a topology on T^*M by using the charts on M, just as we did with the tangent bundle TM. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M. Also adopt the notation $T^*U = \pi^{-1}(U) = \bigsqcup_{p\in U} T_p^*M$. Then we have the bijective map

$$\tilde{\varphi} : T^*U \to \varphi(U) \times \mathbb{R}^n, \qquad \omega_p \mapsto (\varphi(p), a_1(p), \dots, a_n(p)). \tag{3.3}$$

With the standard topology on $\varphi(U) \times \mathbb{R}^n$, we can give T^*U the topology induced by $\tilde{\varphi}$. Then, letting $\mathcal{B}_U$ be the collection of all open sets in T^*U, we can give T^*M the topology generated by $\mathcal{B} = \bigcup_{(U,\varphi)} \mathcal{B}_U$, where the union is taken over all charts (U, φ) on M. The exact same analysis as tangent bundles shows that T^*M is a smooth manifold of dimension $2n$, and the projection $\pi : T^*M \to M$ is a smooth map.

A 1-form ω on M is said to be smooth if ω is a smooth section of π. That is, ω is smooth if $\omega : M \to T^*M$ is a smooth map between smooth manifolds.

Proposition 3.2. *Let ω be a 1-form on a smooth manifold M. Then the following are equivalent.*

1. *ω is smooth.*
2. *For any chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M, the functions $a_i : U \to \mathbb{R}$ defined as $\omega = \sum_{i=1}^n a_i dx^i$ are smooth.*

Let X be a vector field on a smooth manifold M, and let ω be a 1-form on M. Then we can define the map $\omega(X) : M \to \mathbb{R}$ as $\omega(X)(p) = \omega_p(X_p)$ for all $p \in M$. That is, $\omega(X)$ is the function that takes a point $p \in M$ and returns the value of the linear map $\omega_p : T_pM \to \mathbb{R}$ evaluated at the vector $X_p \in T_pM$.

Proposition 3.3. *Let ω be a 1-form on a smooth manifold M. Then ω is a smooth 1-form if and only if $\omega(X)$ is a smooth function on M for all smooth vector fields X on M.*

Proof. Let ω be a smooth 1-form on M. Let X be a smooth vector field on M. Then the map $X : M \to TM$ is smooth, and so the composition $\omega(X) = \omega \circ X : M \to \mathbb{R}$ is smooth. Conversely, suppose that $\omega(X)$ is a smooth function on M for all smooth vector fields X on M. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M. Then the partials $\{\frac{\partial}{\partial x^i}\}_{i=1}^n$ form a basis for the tangent space T_pM for all $p \in U$, and so we can write

$$\omega = \sum_{i=1}^n a_i dx^i \tag{3.4}$$

for some functions $a_i : U \to \mathbb{R}$. Then we have

$$a_i = dx^i \left(\frac{\partial}{\partial x^i} \right) = \omega \left(\frac{\partial}{\partial x^i} \right) \tag{3.5}$$

for all $i = 1, \dots, n$. Since $\frac{\partial}{\partial x^i}$ is a smooth vector field on U, we have that $\omega\left(\frac{\partial}{\partial x^i}\right)$ is a smooth function on U, and so each of the functions a_i are smooth. Thus, ω is a smooth 1-form on M. ■

Recall that $\mathfrak{X}(M)$ is the real vector space of all smooth vector fields on a smooth manifold M, which is also a $C^\infty(M)$-module via $(fX)(p) = f(p)X_p$ for all $f \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, and $p \in M$. Similarly, we have the real vector space $\Omega^1(M)$ of all smooth 1-forms on M, which is also a $C^\infty(M)$-module via $(f\omega)(p) = f(p)\omega_p$ for all $f \in C^\infty(M)$, $\omega \in \Omega^1(M)$, and $p \in M$. Therefore, we can define the map $\Omega^1(M) \times \mathfrak{X}(M) \to C^\infty(M)$ as $(\omega, X) \mapsto \omega(X)$, which is $\mathbb{R}$-bilinear and $C^\infty(M)$-bilinear. That is, we have

$$\begin{aligned} (\omega_1 + \omega_2)(X) = \omega_1(X) + \omega_2(X), \qquad \omega(X_1 + X_2) = \omega(X_1) + \omega(X_2), \\ (f\omega)(X) = f(\omega(X)), \qquad \omega(fX) = f(\omega(X)) \end{aligned} \tag{3.6}$$

Now, if $T : V \to W$ is a linear map between finite-dimensional vector spaces V and W, then the dual map $T^* : W^* \to V^*$ is defined as $T^*(\omega) = \omega \circ T$ for all $\omega \in W^*$. That is, T^* takes a linear functional $\omega : W \to \mathbb{R}$ and returns the linear functional $\omega \circ T : V \to \mathbb{R}$. The dual map is also linear, and satisfies $(S \circ T)^* = T^* \circ S^*$ for all linear maps $T : V \to W$ and $S : W \to U$ between finite-dimensional vector spaces V, W, and U. We can extend this notion to smooth maps between smooth manifolds.

Let $F : N \to M$ be a smooth map between smooth manifolds N and M. Then the differential $F_{*,p} : T_pN \to T_{F(p)}M$ is a linear map between the tangent spaces at $p \in N$ and $F(p) \in M$. Therefore, we can define the dual map $F_p^* : T_{F(p)}^*M \to T_p^*N$ as $F_p^*(\omega) = \omega \circ F_{*,p}$ for all $\omega \in T_{F(p)}^*M$. That is, F_p^* takes a linear functional $\omega_{F(p)} : T_{F(p)}M \to \mathbb{R}$ and returns the linear functional $\omega_{F(p)} \circ F_{*,p} : T_pN \to \mathbb{R}$. The dual map is also linear, and satisfies $(G \circ F)_p^* = F_p^* \circ G_{F(p)}^*$ for all smooth maps $F : N \to M$ and $G : M \to P$ between smooth manifolds N, M, and P. If ω is a 1-form on M, then we can define the pullback of ω by F as the 1-form $F^*\omega$ on N defined as $(F^*\omega)_p = F_p^*(\omega_{F(p)})$ for all $p \in N$.

Proposition 3.4. *Let ω be a smooth 1-form on a smooth manifold M, and let $F : N \to M$ be a smooth map between smooth manifolds N and M. Then the pullback $F^*\omega$ is a smooth 1-form on N.*

Proof. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M, and let $(V, \psi) = (V, y^1, \dots, y^m)$ be a chart on N. Then we have

$$\omega = \sum_{i=1}^n a_i dx^i, \qquad F^*\omega = \sum_{j=1}^m b_j dy^j \tag{3.7}$$

for some smooth functions $a_i : U \to \mathbb{R}$ and $b_j : V \to \mathbb{R}$. Then we have

$$b_j(p) = (F^*\omega)_p\left(\frac{\partial}{\partial y^j}\right) = F_p^*(\omega_{F(p)})\left(\frac{\partial}{\partial y^j}\right) = \omega_{F(p)}(F_{*,p}\left(\frac{\partial}{\partial y^j}\right)) = \sum_{i=1}^{n} a_i(F(p))dx^i(F_{*,p}\left(\frac{\partial}{\partial y^j}\right)). \tag{3.8}$$

Since $F_{*,p}\left(\frac{\partial}{\partial y^j}\right)$ is a smooth vector field on N, we have that $dx^i(F_{*,p}\left(\frac{\partial}{\partial y^j}\right))$ is a smooth function on N. Therefore, each of the functions b_j are smooth, and so $F^*\omega$ is a smooth 1-form on N. ■

3.1.1 k-forms

October 7th.

Moving on to k-forms, recall that $A_k(V)$ is the space of all alternating k-linear maps $\omega : V^k \to \mathbb{R}$ on a finite-dimensional vector space V. It is of dimension $\binom{n}{k}$, and if $\{\alpha_1, \ldots, \alpha_n\}$ is a basis for V^*, then the wedge products $\{\alpha_{i_1} \wedge \cdots \wedge \alpha_{i_k} \mid 1 \le i_1 < \cdots < i_k \le n\}$ form a basis for $A_k(V)$. We can extend this notion to smooth manifolds.

Let M be a smooth manifold of dimension n, and let $p \in M$. We have

$$A_k(T_pM) := \{\omega_p : T_pM^k \to \mathbb{R} \mid \omega_p \text{ is alternating and } k\text{-linear}\} = \bigwedge^k T_p^*M. \tag{3.9}$$

A k-form on M is a map

$$\omega : M \to \bigsqcup_{p\in M} A_k(T_pM) = \bigsqcup_{p\in M} \bigwedge^k T_p^*M \tag{3.10}$$

such that $\omega(p) \in A_k(T_pM)$ for all $p \in M$. If we denote $\mathcal{I}_{k,n} = \{(i_1, \ldots, i_k) \mid 1 \le i_1 < \cdots < i_k \le n\}$, then we have that $\dim \bigwedge^k T_p^*M = \binom{n}{k}$, and if $(U, \varphi) = (U, x^1, \ldots, x^n)$ is a chart on M, then the wedge products $\{(dx^I)_p = (dx^{i_1})_p \wedge \cdots \wedge (dx^{i_k})_p \mid (i_1, \ldots, i_k) \in \mathcal{I}_{k,n}\}$ form a basis for $\bigwedge^k T_p^*M$. Thus, any $\omega_p \in A_k(T_pM)$ can be written as

$$\omega_p = \sum_{I\in\mathcal{I}_{k,n}} a_I(p)(dx^I)_p \tag{3.11}$$

where $a_I(p)$'s are real numbers. More generally, any k-form ω on U can be written as

$$\omega = \sum_{I\in\mathcal{I}_{k,n}} a_I dx^I \tag{3.12}$$

where $a_I : U \to \mathbb{R}$ are real-valued functions on U. If $f_1, \ldots, f_k \in C^\infty(U)$ are smooth functions on U, then we have

$$(df_1 \wedge \cdots \wedge df_k)\left(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}\right) = \det\left(\frac{\partial f_j}{\partial x^{i_l}}\right)_{1\le j,l\le k}. \tag{3.13}$$

We also have the notion of bundles of k-forms on M. The case $k = 1$ gives the cotangent bundle T^*M. These bundles are defined as

$$\bigwedge^k T^*M = \bigsqcup_{p\in M} \bigwedge^k T_p^*M = \bigsqcup_{p\in M} A_k(T_pM) \tag{3.14}$$

with projection $\pi : \bigwedge^k T^*M \to M$ defined as $\pi(\omega_p) = p$ for all $\omega_p \in \bigwedge^k T_p^*M$. A k-form on M, then, is a section of the bundle $\bigwedge^k T^*M$. To induce a topology on $\bigwedge^k T^*M$, we can use the charts on M as we did with the tangent and cotangent bundles. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M. Also adopt the notation $\bigwedge^k T^*U = \pi^{-1}(U) = \bigsqcup_{p\in U} \bigwedge^k T_p^*M$. Then we have the bijective map

$$\tilde{\varphi} : \bigwedge^k T^*U \to \varphi(U) \times \mathbb{R}^{\binom{n}{k}}, \qquad \omega_p \mapsto (\varphi(p), a_{I_1}(p), \ldots, a_{I_{\binom{n}{k}}}(p)). \tag{3.15}$$

With the standard topology on $\varphi(U) \times \mathbb{R}^{\binom{n}{k}}$, we can give $\bigwedge^k T^*U$ the topology induced by $\tilde{\varphi}$. Then, letting $\mathcal{B}_U$ be the collection of all open sets in $\bigwedge^k T^*U$, we can give $\bigwedge^k T^*M$ the topology generated by $\mathcal{B} = \bigcup_{(U,\varphi)} \mathcal{B}_U$, where the union is taken over all charts (U, φ) on M. The exact same analysis as tangent bundles shows that $\bigwedge^k T^*M$ is a smooth manifold of dimension $n + \binom{n}{k}$, and the projection $\pi : \bigwedge^k T^*M \to M$ is a smooth map. The case for $k = 0$ is trivial, as $\bigwedge^0 T^*M = M \times \mathbb{R}$ is a smooth manifold of dimension $n + 1$, and the projection $\pi : \bigwedge^0 T^*M \to M$ is the projection onto the first factor. A smooth k-form ω on M is a smooth section of π. That is, ω is smooth if $\omega : M \to \bigwedge^k T^*M$ is a smooth map between smooth manifolds. Also, a k-form ω on M is smooth if and only if for any chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M, the functions $a_I : U \to \mathbb{R}$ defined as $\omega = \sum_{I \in \mathcal{I}_{k,n}} a_I dx^I$ are smooth. Again, for $k = 0$, a smooth section of $\pi : M \times \mathbb{R} \to M$ is a smooth map $f : M \to M \times \mathbb{R}$ such that $\pi(f(p)) = p$ for all $p \in M$. That is, $f(p) = (p, g(p))$ for some smooth function $g : M \to \mathbb{R}$. Thus, a smooth 0-form on M is simply a smooth function on M.

October 9th.

Same as before, given a smooth map $F : N \to M$ between smooth manifolds N and M, we can define the pullback of a k-form ω on M by F as the k-form $F^*\omega$ on N defined as $(F^*\omega)_p = F_p^*(\omega_{F(p)})$ for all $p \in N$. That is, $(F^*\omega)_p : T_pN^k \to \mathbb{R}$ is the alternating k-linear map defined as

$$(F^*\omega)_p(X_1, \ldots, X_k) = \omega_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \tag{3.16}$$

for all $X_1, \ldots, X_k \in T_pN$. The pullback of a k-form by a smooth map is also smooth. That is, if ω is a smooth k-form on M, then $F^*\omega$ is a smooth k-form on N. The proof is the same as the case for 1-forms.

From here, we define

$$\Omega^k(M) = \{\omega : M \to \bigwedge^k T^*M \mid \omega \text{ is smooth}\}, \tag{3.17}$$

the vector space of all smooth k-forms on M, which is also a $C^\infty(M)$-module via $(f\omega)(p) = f(p)\omega_p$ for all $f \in C^\infty(M)$, $\omega \in \Omega^k(M)$, and $p \in M$.

October 12th.

The wedge product of forms is also defined here, as follows. Let $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$ be k- and l-forms on a smooth manifold M. Then we can define the wedge product $\omega \wedge \eta$ as the $(k + l)$-form on M defined as

$$(\omega \wedge \eta)_p = \omega_p \wedge \eta_p = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \omega_p(X_{\sigma(1)}, \ldots, X_{\sigma(k)}) \eta_p(X_{\sigma(k+1)}, \ldots, X_{\sigma(k+l)}) \tag{3.18}$$

for all $p \in M$ and $X_1, \ldots, X_{k+l} \in T_pM$.

Proposition 3.5. *If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$ are smooth k- and l-forms on a smooth manifold M, then $\omega \wedge \eta$ is a smooth $(k + l)$-form on M.*

Proof. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M. Then we have

$$\omega = \sum_{I \in \mathcal{I}_{k,n}} a_I dx^I, \qquad \eta = \sum_{J \in \mathcal{I}_{l,n}} b_J dx^J \tag{3.19}$$

for some smooth functions $a_I : U \to \mathbb{R}$ and $b_J : U \to \mathbb{R}$. Then we have

$$(\omega \wedge \eta) = \sum_{I,J} a_I b_J dx^I \wedge dx^J. \tag{3.20}$$

Since the product of smooth functions is smooth, we have that each of the functions $a_I b_J$ are smooth. Thus, $\omega \wedge \eta$ is a smooth $(k + l)$-form on M. ∎

Again, the wedge product is $\mathbb{R}$-bilinear, $C^\infty(M)$-bilinear, associative, and anti-commutative. From here, for a smooth manifold M of dimension n, we define

$$\Omega^*(M) = \bigoplus_{k=0}^{n} \Omega^k(M) = \bigoplus_{k=0}^{\infty} \Omega^k(M), \tag{3.21}$$

which is a graded $\mathbb{R}$-algebra with respect to the wedge product. A smooth map $F : N \to M$ between smooth manifolds N and M, then, induces a graded $\mathbb{R}$-algebra homomorphism $F^* : \Omega^*(M) \to \Omega^*(N)$. We first show the pullback for general k-forms. For a k-form ω on M, we have

$$(F^*\omega)_p(X_1, \ldots, X_k) = \omega_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \tag{3.22}$$

for all $p \in N$ and $X_1, \ldots, X_k \in T_pN$. Then we have

$$\begin{aligned}(F^*(\omega \wedge \eta))_p(X_1, \ldots, X_{k+l}) &= (\omega \wedge \eta)_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_{k+l})) \\ &= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \omega_{F(p)}(F_{*,p}(X_{\sigma(1)}), \ldots, F_{*,p}(X_{\sigma(k)})) \\ &\qquad \cdot \eta_{F(p)}(F_{*,p}(X_{\sigma(k+1)}), \ldots, F_{*,p}(X_{\sigma(k+l)})) \\ &= (F^*\omega \wedge F^*\eta)_p(X_1, \ldots, X_{k+l})\end{aligned} \tag{3.23}$$

and

$$\begin{aligned}(F^*(f\omega + g\eta))_p(X_1, \ldots, X_k) &= (f\omega + g\eta)_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \\ &= f(F(p))\omega_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) + g(F(p))\eta_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \\ &= (F^*f)(p)(F^*\omega)_p(X_1, \ldots, X_k) + (F^*g)(p)(F^*\eta)_p(X_1, \ldots, X_k) \\ &= ((F^*f)(F^*\omega) + (F^*g)(F^*\eta))_p(X_1, \ldots, X_k)\end{aligned} \tag{3.24}$$

showing that $F^*(\omega \wedge \eta) = F^*\omega \wedge F^*\eta$ and $F^*(f\omega + g\eta) = (F^*f)(F^*\omega) + (F^*g)(F^*\eta)$ for all $\omega, \eta \in \Omega^*(M)$ and $f, g \in C^\infty(M)$. Now, we show that if $\omega \in \Omega^k(M)$, then $F^*\omega \in \Omega^k(N)$. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M, and let $(V, \psi) = (V, y^1, \ldots, y^m)$ be a chart on N, with $F(U) \subseteq V$. Then $F^*\omega$ is smooth since

$$\begin{aligned}(F^*\omega)_p = F_p^*(\omega_{F(p)}) &= \sum_{I \in \mathcal{I}_{k,n}} a_I(F(p)) F_p^*(dx^I) = \sum_{I \in \mathcal{I}_{k,n}} a_I(F(p)) d(x^{i_1} \circ F)_p \wedge \cdots \wedge d(x^{i_k} \circ F)_p \\ &= \sum_{I \in \mathcal{I}_{k,n}} a_I(F(p)) d(F^*x^{i_1})_p \wedge \cdots \wedge d(F^*x^{i_k})_p\end{aligned} \tag{3.25}$$

where each function $a_I \circ F$ and each function $x^i \circ F$ is smooth on $F^{-1}(U)$. Hence, $F^*\omega$ is a smooth k-form on N.

3.1.2 Exterior Differentiation

Let $A = \bigoplus_{n=0}^{\infty} A_n$ be a graded algebra over $\mathbb{R}$. An $\mathbb{R}$-linear map $D : A \to A$ is said to be an antiderivation of degree $+1$ if $D(ab) = (Da)b + (-1)^{\deg a} a(Db)$ for all $a \in A_k$ and $b \in A_l$, and $Da \in A_{k+1}$ whenever $a \in A_k$. In particular, let $A = \Omega^*(M)$ be the graded algebra of smooth differential forms on a smooth manifold M, and define the exterior derivative $d : \Omega^*(M) \to \Omega^*(M)$ as the $\mathbb{R}$-linear map defined as

$$d\left(\sum a_I dx^I\right) = \sum da_I \wedge dx^I \tag{3.26}$$

after coordinates are chosen. Then d is an antiderivation of degree $+1$, satisfies $d^2 = 0$, and for any $f \in C^\infty(M)$, $df \in \Omega^1(M)$ is the differential of f. Moreover, if D is any antiderivation of degree $+1$ on $\Omega^*(M)$ such that $D^2 = 0$ and $Df = df$ for all $f \in C^\infty(M)$, then $D = d$. That is, the exterior derivative is the unique antiderivation of degree $+1$ on $\Omega^*(M)$ that squares to zero and agrees with the differential on smooth functions.

3.2 Orientable Manifolds

October 14th.

We first provide the theory for a finite dimensional vector space V over $\mathbb{R}$, of dimension n. On the collection of all ordered bases of V, we define an equivalence relation as $(v_1, \dots, v_n) \sim (w_1, \dots, w_n)$ if and only if the change of basis matrix from $(v_1, \dots, v_n)$ to $(w_1, \dots, w_n)$ has a positive determinant. Reflexivity, symmetry, and transitivity may be easily verified. This equivalence relation on the set of ordered basis of V partitions it into 2 equivalence classes. An orientation on V is a choice between these 2 equivalence clases or, equivalently, a choice of an ordered basis and its corresponding equivalence class. We say V is oriented after making such a choice. For a finite dimensional vector space V over $\mathbb{R}$, we shall use μ and $-\mu$ to denote the two orientations.

Lifting this theory, let M be a smooth manifold of dimension n. Recall that a vector field X is a section of the tangent bundle TM such that $X : M \to TM$ satisfies $\pi \circ X = \mathrm{id}_M$, and a(n ordered) frame $(X_1, X_2, \dots, X_n)$ on M is a collection of vector fields on M such that for all $p \in M$, $(X_{1,p}, X_{2,p}, \dots, X_{n,p})$ is a(n ordered) basis for T_pM.

Definition 3.6. A *pointwise orientation* μ on a smooth manifold M is a choice of orientation μ_p on T_pM for all $p \in M$. It is said to be continuous at a point $p \in M$ if p has a neighbourhood U on which μ is represented by a continuous frame on U; that is, there exist continuous vector fields $X_1, \dots, X_n$ on U such that for all $q \in U$, $[(X_{1,p}, X_{2,p}, \dots, X_{n,p})] = \mu_p$.
An *orientation* on M is a pointwise orientation μ that is continuous for all $p \in M$. A manifold on which there exists an orientation is called an *orientable manifold*. A manifold with a choice of orientation is called a *oriented manifold*.

Proposition 3.7. *Let M be a smooth orientable connected manifold. Then M has exactly 2 orientations.*

Proof. Take two orientations μ, ν on M. Define the map $f : M \to \{\pm 1\}$ as $f(p) = 1$ if $\mu_p = \nu_p$ and $f(p) = -1$ if $\mu_p \neq \nu_p$. We claim that f is a continuous map. Let $p \in U$, where U is a connected neighbourhood of p. Then continuity at p gives $\mu = (X_1, \dots, X_n)$ and $\nu = (Y_1, \dots, Y_n)$, two collections of continuous vector fields on U which form a frame. Thus, there exists a matrix of continuous functions $A : U \to M_n(\mathbb{R})$ such that

$$(Y_{1,q}, \dots, Y_{n,q}) = (X_{1,q}, \dots, X_{n,q})A(q) \qquad \text{for all } q \in U. \tag{3.27}$$

This implies that the map $\det(A(\cdot)) : U \to \mathbb{R}^\times$ is a continuous function; so it must be positive on the entirety of U, or negative. Moreover, $f : U \to \{\pm 1\}$ may be written as $f(q) = \mathrm{sgn}(\det(A(q)))$ for all $q \in U$, showing that f is locally constant. Since M is connected, f is constant, finally implying that $\mu = \nu$ or $\mu = -\nu$. ∎

October 16th.

Proposition 3.8. *Let M be a smooth manifold of dimension n. A frame $(X_1, \dots, X_n)$ on M gives a pointwise continuous orientation on M if and only if for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M with $p \in U$ such that $(dx^1 \wedge \cdots \wedge dx^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$.*

Proof. Let $(X_1, \dots, X_n)$ be a frame on M which gives a pointwise continuous orientation on M. Let $p \in M$. Then there exists a neighbourhood U of p and a continuous frame $(Y_1, \dots, Y_n)$ on U such that for all $q \in U$, $[(Y_{1,q}, \dots, Y_{n,q})] = [(X_{1,q}, \dots, X_{n,q})]$. $[(\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q]$ is an orientation on T_qM for all $q \in U$. Since $Y_1, \dots, Y_n$ are continuous vector fields on U, the coordinate functions $a_{ij} : U \to \mathbb{R}$ defined as $Y_i = \sum_{j=1}^n a_{ij}(\frac{d}{dx^j})$ are continuous. Thus, the map $\det(a_{ij}) : U \to \mathbb{R}^\times$ is continuous, and so it must be positive on the entirety of U, or negative. If this map is negative, we can replace Y_1 with $-Y_1$ to make it positive. Then we have

$$(dx^1 \wedge \cdots \wedge dx^n)_q(Y_{1,q}, \dots, Y_{n,q}) = \det(a_{ij}(q)) > 0. \tag{3.28}$$

Finally, since $[(Y_{1,q}, \dots, Y_{n,q})] = [(X_{1,q}, \dots, X_{n,q})]$, we have that

$$(dx^1 \wedge \cdots \wedge dx^n)_q(X_{1,q}, \dots, X_{n,q}) > 0. \tag{3.29}$$

For the converse, let $(X_1, \dots, X_n)$ be a frame on M such that for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M with $p \in U$ such that $(dx^1 \wedge \cdots \wedge dx^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$. Then for all $q \in U$, we have

$$(dx^1 \wedge \cdots \wedge dx^n)_q(X_{1,q}, \dots, X_{n,q}) > 0. \tag{3.30}$$

We show that $[(X_{1,q}, \dots, X_{n,q})] = [(\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q]$ for all $q \in U$. Let $A(q)$ be the change of basis matrix from $(X_{1,q}, \dots, X_{n,q})$ to $((\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q)$. Then we have

$$(dx^1 \wedge \cdots \wedge dx^n)_q(X_{1,q}, \dots, X_{n,q}) = \det(A(q)) > 0. \tag{3.31}$$

This shows that the frame $(X_{1,q}, \dots, X_{n,q})$ is positively oriented with respect to the coordinate frame $((\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q)$ for all $q \in U$, and hence the frame $(X_1, \dots, X_n)$ gives a pointwise continuous orientation on M. ■

Proposition 3.9. *Let M be a smooth manifold of dimension n. Then M is orientable if and only if there exists a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$.*

Proof. Suppose M is orientable, and $X_1, \dots, X_n$ is a frame on M which gives a pointwise continuous orientation on M. Then for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M with $p \in U$ such that $(dx^1 \wedge \cdots \wedge dx^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$, or on U, $(dx^1 \wedge \cdots \wedge dx^n)(X_1, \dots, X_n) > 0$. Vary $p \in M$ to get an open cover $\{U_\alpha\}$ of M with charts $(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \dots, x^n_\alpha)$ on M such that $(dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha)(X_1, \dots, X_n) > 0$ on U_α. Let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to the open cover $\{U_\alpha\}$. Then we can define a smooth n-form $\omega \in \Omega^n(M)$ as

$$\omega = \sum_\alpha \rho_\alpha dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha. \tag{3.32}$$

Then for all $p \in M$, we have

$$\omega_p(X_{1,p}, \dots, X_{n,p}) = \sum_\alpha \rho_\alpha(p)(dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha)_p(X_{1,p}, \dots, X_{n,p}) > 0. \tag{3.33}$$

This shows that ω is a smooth nowhere vanishing n-form on M. For the converse, suppose there exists a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$. Then for all $p \in M$, we have $\omega_p \neq 0$, and hence there exists a basis $(X_{1,p}, \dots, X_{n,p})$ for T_pM such that $\omega_p(X_{1,p}, \dots, X_{n,p}) > 0$. Vary $p \in M$ to get a frame $(X_1, \dots, X_n)$ on M such that $\omega(X_1, \dots, X_n) > 0$. Define an orientation on T_pM by declaring an ordered basis $(v_1, \dots, v_n)$ to be positive if and only if

$$\omega_p(v_1, \dots, v_n) > 0. \tag{3.34}$$

This is well-defined, since replacing a basis by another one multiplies the value of ω_p by the determinant of the change-of-basis matrix. It remains to verify continuity. Let $p \in M$ and choose a chart $(U, x^1, \dots, x^n)$ around p. Since ω is nowhere vanishing, we can write

$$\omega = f\, dx^1 \wedge \cdots \wedge dx^n \tag{3.35}$$

on U, where $f : U \to \mathbb{R}$ is smooth and nowhere zero. After shrinking U if necessary, f has constant sign. Consequently, the coordinate frame $(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$, or the same frame with one vector multiplied by -1, represents the orientation defined above throughout U. Thus the pointwise orientation is continuous, so M is orientable. ■

October 21st.

Definition 3.10. Let $\{(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \dots, x^n_\alpha)\}_{\alpha \in \Lambda}$ be an atlas on a smooth manifold M of dimension n. It is termed an *orientable atlas* if for all $\alpha, \beta \in \Lambda$ with $U_\alpha \cap U_\beta \neq \emptyset$, the Jacobian determinant of the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on $\varphi_\alpha(U_\alpha \cap U_\beta)$; that is,

$$\det\left(\frac{\partial x^i_\beta}{\partial x^j_\alpha}\right)_{1 \le i,j \le n} > 0 \text{ on } \varphi_\alpha(U_\alpha \cap U_\beta). \tag{3.36}$$

Proposition 3.11. *Let M be a smooth manifold of dimension n. Then M is orientable if and only if there exists an orientable atlas on M.*

Proof. If M is orientable, then there exists a frame $(X_1, \dots, X_n)$ on M such that there exist charts $(U, \varphi) = (U, x^1, \dots, x^n)$ covering M satisfying $(dx^1 \wedge \cdots \wedge dx^n)(X_1, \dots, X_n) > 0$. Let $\{(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \dots, x^n_\alpha)\}_{\alpha \in \Lambda}$ be this atlas of charts on M. Then for all $\alpha, \beta \in \Lambda$ with $U_\alpha \cap U_\beta \neq \emptyset$, we have

$$\det\left(\frac{\partial x^i_\beta}{\partial x^j_\alpha}\right) \cdot (dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha)(X_1, \dots, X_n) = (dx^1_\beta \wedge \cdots \wedge dx^n_\beta)(X_1, \dots, X_n) > 0 \tag{3.37}$$

implying that the Jacobian determinant of the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on $\varphi_\alpha(U_\alpha \cap U_\beta)$. Thus, $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ is an orientable atlas on M. Conversely, if there exists an orientable atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ on M, let μ be the pointwise orientation on M defined as follows. For all $p \in M$, choose $\alpha \in \Lambda$ such that $p \in U_\alpha$, and define $\mu_p = [((\frac{\partial}{\partial x^1_\alpha})_p, \dots, (\frac{\partial}{\partial x^n_\alpha})_p)]$. This is well-defined, since if $p \in U_\alpha \cap U_\beta$, then the Jacobian determinant of the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on $\varphi_\alpha(U_\alpha \cap U_\beta)$, and hence

$$\left[\left(\left(\frac{\partial}{\partial x^1_\alpha}\right)_p, \dots, \left(\frac{\partial}{\partial x^n_\alpha}\right)_p\right)\right] = \left[\left(\left(\frac{\partial}{\partial x^1_\beta}\right)_p, \dots, \left(\frac{\partial}{\partial x^n_\beta}\right)_p\right)\right]. \tag{3.38}$$

■

Let M be a connected smooth manifold of dimension n. Then M is orientable if and only if there exists a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$; on the set of all smooth nowhere vanishing n-forms on M, we define an equivalence relation as $\omega \sim \eta$ if and only if $\omega = f\eta$ for some smooth function $f : M \to \mathbb{R}$ such that $f(p) > 0$ for all $p \in M$. This equivalence relation partitions the set of all smooth nowhere vanishing n-forms on M into 2 equivalence classes. Thus, an orientation determined by a smooth nowhere vanishing n-form ω on M is exactly the orientation given by a frame $(X_1, \dots, X_n)$ on M such that $\omega(X_1, \dots, X_n) > 0$. In particular, if ω and η are smooth nowhere vanishing n-forms on M, then they determine the same orientation on M if and only if $\omega = f\eta$ for some smooth function $f > 0$.

3.2.1 Manifolds with Boundary

October 24th.

Hereforth, we use $\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n \geq 0\}$ to denote the closed upper half-space in $\mathbb{R}^n$.

Definition 3.12. A topological space X is *locally* $\mathbb{H}^n$ if for all $p \in X$, there exists a neighbourhood U of p and a homeomorphism $\varphi : U \to V$ onto an open subset V of $\mathbb{H}^n$ such that $\varphi(p) = 0$. M is termed a *topological manifold with boundary* of dimension n if it is Hausdorff, second countable, and locally $\mathbb{H}^n$.

A smooth atlas on M is a collection of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that for all $\alpha, \beta \in \Lambda$ with $U_\alpha \cap U_\beta \neq \emptyset$, the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is smooth on $\varphi_\alpha(U_\alpha \cap U_\beta)$. A smooth manifold with boundary is a topological manifold with boundary equipped with a smooth atlas.

For the closed half space $\mathbb{H}^n$, we define

$$\partial\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n = 0\}, \qquad (\mathbb{H}^n)^\circ = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n > 0\} \tag{3.39}$$

so that $\mathbb{H}^n = (\mathbb{H}^n)^\circ \sqcup \partial\mathbb{H}^n$. Note that if U and V are open subsets of $\mathbb{H}^n$ and $f : U \to V$ is a diffeomorphism, then $f(\partial\mathbb{H}^n \cap U) \subseteq \partial\mathbb{H}^n \cap V$ and $f((\mathbb{H}^n)^\circ \cap U) \subseteq (\mathbb{H}^n)^\circ \cap V$. Here, $\partial\mathbb{H}^n$ denotes the boundary of $\mathbb{H}^n$, and $(\mathbb{H}^n)^\circ$ denotes the interior of $\mathbb{H}^n$. One can show that $\partial\mathbb{H}^n$ is a smooth manifold of dimension $n-1$, and $(\mathbb{H}^n)^\circ$ is a smooth manifold of dimension n. This can be extended to any smooth manifold with boundary M. We denote

$$\partial M = \{p \in M \mid \text{there exists a chart } (U, \varphi) \text{ on } M \text{ with } p \in U \text{ and } \varphi(p) \in \partial\mathbb{H}^n\}, \qquad M^\circ = M \setminus \partial M. \tag{3.40}$$

One can show that ∂M is a smooth manifold of dimension $n-1$, and M° is a smooth manifold of dimension n. We call ∂M the *boundary* of M, and M° the *interior* of M. Note that if M is a smooth manifold with boundary, then ∂M is a smooth manifold without boundary.

Smooth functions on a smooth manifold with boundary M are defined as follows. A function $f : M \to \mathbb{R}$ is smooth if for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M with $p \in U$ such that $f \circ \varphi^{-1}$ is smooth on $\varphi(U)$. Smooth maps between smooth manifolds with boundary are defined similarly. Germs of smooth functions on a smooth manifold with boundary M are defined just as for smooth manifolds without boundary. The tangent space T_pM at a point $p \in M$ is defined as the set of derivations on the germs of smooth functions at p. We may then, analogously, define

$$T_p^*M = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R}), \qquad \bigwedge^k T_p^*M = A_k(T_pM) \tag{3.41}$$

and

$$TM = \bigsqcup_{p \in M} T_pM, \qquad T^*M = \bigsqcup_{p \in M} T_p^*M. \tag{3.42}$$

A smooth k-form on a smooth manifold with boundary M is defined as a smooth section of the bundle $\bigwedge^k T^*M$. An orientation on a smooth manifold with boundary M, then, is one of three equivalent definitions: a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$, a pointwise continuous frame $(X_1, \ldots, X_n)$ on M, or an orientable atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ on M.

Let $\beta \in \bigwedge^k V^* = A_k(V)$ be a k-covector on a finite dimensional vector space V over $\mathbb{R}$, where $\dim V \geq k \geq 2$. If a vector $v \in V$ is fixed, then we can define a $(k-1)$-covector $i_v\beta \in \bigwedge^{k-1} V^*$ as

$$(i_v\beta)(v_1, \ldots, v_{k-1}) = \beta(v, v_1, \ldots, v_{k-1}). \tag{3.43}$$

If $k = 1$, then we define $i_v\beta \in \bigwedge^0 V^* = \mathbb{R}$ as $(i_v\beta) = \beta(v)$. $k = 0$ admits $i_v\beta = 0$. Moreover, if $\beta^1, \ldots, \beta^k \in V^*$ are 1-covectors on V, then $\beta^1 \wedge \cdots \wedge \beta^k \in \bigwedge^k V^*$ is a k-covector on V, and we have

$$i_v(\beta^1 \wedge \cdots \wedge \beta^k) = \sum_{i=1}^{k} (-1)^{i-1} \beta^1 \wedge \cdots \wedge \beta^{i-1} \wedge \beta^{i+1} \wedge \cdots \wedge \beta^k. \tag{3.44}$$

The same construction can be lifted to a smooth manifold, M. Let $\omega \in \Omega^k(M)$ be a smooth k-form on M, and let $X \in \mathfrak{X}(M)$ be a vector field on M. Then we can define a smooth $(k-1)$-form $i_X\omega \in \Omega^{k-1}(M)$ as

$$(i_X\omega)_p = i_{X_p}(\omega_p) \tag{3.45}$$

for all $p \in M$.

Note that if M is an oriented smooth manifold with boundary, then ∂M is an oriented smooth manifold without boundary.

Definition 3.13. Let M be a smooth manifold with boundary. Suppose $p \in \partial M$. A vector $X_p \in T_pM$ is said to be *inward pointing* if

1. $X_p \notin T_p(\partial M)$, and
2. there exists a smooth map $c : [0, \epsilon) \to M$ for some $\epsilon > 0$ such that $c(0) = p$, $c'(0) = X_p$, and $c((0, \epsilon)) \subseteq M^\circ$.

It is said to be *outward pointing* if $-X_p$ is inward pointing.

Definition 3.14. For a smooth manifold with boundary M, a vector field X along the boundary ∂M is a function $X : \partial M \to \bigsqcup_{p \in \partial M} T_pM$ such that $X(p) \in T_pM$ for all $p \in \partial M$. Such a vector field along the boundary is said to be smooth at $p \in \partial M$ if for any chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M with $p \in U$, X when written as $X(q) = \sum_{i=1}^n a^i(q)(\frac{\partial}{\partial x^i})_q$ for $q \in U \cap \partial M$ has smooth coefficients $a^i : U \cap \partial M \to \mathbb{R}$. A vector field along the boundary is said to be smooth if it is smooth at all points $p \in \partial M$. It is outward pointing at $p \in \partial M$ if and only if $a^n(p) > 0$ in any chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M with $p \in U$.

Proposition 3.15. *For any smooth manifold with boundary, there exists a smooth outward pointing vector field along the boundary.*

Proof. Let $p \in \partial M$, and choose a chart $(U_\alpha, x^1_\alpha, \dots, x^n_\alpha)$ about p. Define $X_\alpha = -\frac{\partial}{\partial x^n_\alpha}$ on $U_\alpha \cap \partial M$. Then X_α is a smooth outward pointing vector field along the boundary on $U_\alpha \cap \partial M$. Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ be an open cover of ∂M by such charts, and let $\{\rho_\alpha\}_{\alpha\in\Lambda}$ be a smooth partition of unity subordinate to this open cover. Then we can define a smooth outward pointing vector field along the boundary as

$$X = \sum_{\alpha\in\Lambda} \rho_\alpha X_\alpha. \tag{3.46}$$

This is well-defined and smooth, and it is outward pointing at each point of ∂M. ∎

Now, let M be an oriented smooth manifold with boundary of dimension n. Then ∂M is an oriented smooth manifold without boundary of dimension $n-1$: let $\omega \in \Omega^n(M)$ be a smooth nowhere vanishing n-form on M which determines the orientation on M. Let X be a smooth outward pointing vector field along the boundary ∂M, via the above proposition. Then we can define a smooth nowhere vanishing $(n-1)$-form $\eta \in \Omega^{n-1}(\partial M)$ on ∂M as

$$\eta = i_X \omega|_{\partial M}. \tag{3.47}$$

This η cannot be zero; if it were at some point $p \in \partial M$, then $\omega_p(X_p, v_1, \dots, v_{n-1}) = 0$ for all $v_1, \dots, v_{n-1} \in T_p(\partial M)$. But since X_p is outward pointing, it is not in $T_p(\partial M)$, and hence $(X_p, v_1, \dots, v_{n-1})$ forms a basis for T_pM. This contradicts the fact that ω is nowhere vanishing. Thus, η is a smooth nowhere vanishing $(n-1)$-form on ∂M, and it determines an orientation on ∂M. Moreover, an ordered basis $(v_1, v_2, \dots, v_{n-1})$ for $T_p(\partial M)$ is positively oriented with respect to the orientation on ∂M if and only if $(X_p, v_1, v_2, \dots, v_{n-1})$ is positively oriented with respect to the orientation on M.

Index

C^∞-compatible, 9
C^k-continuous, 1
F-related, 28
k-linear, 3
k-tensors, 4
n-sphere, 13
1-form, 35

algebra, 1
alternating k-linear form, 4
alternating k-tensors, 4
alternation, 4
antiderivation, 9
atlas, 10

base space, 22
bilinear form, 3
boundary, 42
bump function, 24
bundle map, 23

closed k-form, 9
codimension, 17
complete vector field, 27
coordinate chart, 9
cotangent bundle, 35
cotangent space, 7
critical point, 17
critical value, 17

de Rham cohomology, 9
derivation, 3, 7
diffeomorphism, 12
differential k-form, 7
differential 1-form, 7
directional derivative, 2
dual basis, 3
dual space, 3

embedding, 24
exact k-form, 9
exterior algebra, 8
exterior derivative, 8

frame, 23

germ, 2
global flow, 27
Grassmannian space, 13

immersed submanifold, 23
immersion, 16
integral curve, 26
interior, 42
invariance of dimension, 9
inward pointing, 43

Jacobian matrix, 15, 16

left-invariant vector field, 31
Lie algebra, 28
Lie group, 29
Lie subgroup, 29
local flow, 27
locally $\mathbb{H}^n$, 42
locally euclidean, 9
locally finite, 24

map of Lie groups, 29
maximal atlas, 10
maximal integral curve, 26
morphism, 22

orientable atlas, 41
orientable manifold, 40
orientation, 40
oriented manifold, 40
outward pointing, 43

partition of unity, 24
pointwise orientation, 40

real analytic, 1
real projective space, 13
regular point, 17
regular submanifold, 17
regular value, 17

section, 23
smooth, 1
smooth curve, 16
smooth manifold, 10
smooth vector bundle, 22

smooth vector field, 7
star-shaped, 2
submersion, 16
subordinate, 25
symmetric k-linear form, 4
symmetric k-tensors, 4
symmetrization, 4

tangent bundle, 21
tangent space, 6, 15
Taylor's theorem, 2
tensor product, 4
topological manifold, 9
topological manifold with boundary, 42
total space, 22
trivial vector bundle, 23

vector field, 7, 22

wedge product, 4